KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS



Hospital Chatbot for Color-Coded Clinical Pathways Using Bidirectional Encoder Representations from Transformers

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BY

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DECLARATION

We hereby declare that this submission is our own work towards the award of the Bachelor of Science degree in Mathematics and that, to the best of our knowledge, it contains no material previously published by another person nor material which has been accepted for the award of any other degree of the university, except where due acknowledgment has been made in the text.

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DEDICATION

To my family, lecturers, and the patients whose care motivates this work.

ABSTRACT

Effective triage is critical for matching hospital resources to patient acuity. In low- and middle-income countries, including Ghana, hospitals frequently suffer from flow management failures characterized by overcrowding, delayed initial assessments, and acuity discontinuity. To address these operational bottlenecks, this project presents a hospital chatbot powered by a fine-tuned Biomedical Bidirectional Encoder Representations from Transformers (BioBERT) model. The system estimates Emergency Severity Index (ESI)–style urgency from unstructured chief complaints and applies a proposed deterministic map f_{SATS} to generate SATS-aligned colour and pathway recommendations.

The underlying mathematical architecture uses tokenisation, continuous vector embeddings, and multi-head self-attention to generate contextualized text summaries, which are passed through a Softmax classification head to predict a categorical distribution over five mutually exclusive ESI-style urgency levels. The heuristic f_{SATS} then translates these predictions into triage routing colours (Red, Orange, Yellow, or Green); it is an operational crosswalk, not a claim of clinical ESI–SATS equivalence. To support clinical safety, the pipeline features a mathematical relevance gate to filter non-medical inputs prior to classification. Furthermore, the model acts as a text-first precursor to the standard Triage Early Warning Score (TEWS), supplying an initial pathway card $P(C)$, detailing destination and maximum wait time, before physical vitals can be obtained.

Primary reported metrics are derived from external evaluation on the Medical Information Mart for Intensive Care IV Emergency Department module (MIMIC-IV-ED) and the National Hospital Ambulatory Medical Care Survey (NHAMCS) 2018–22, providing evaluation on real-world US emergency-department phrasing rather than synthetic holdouts. On the NHAMCS evaluation set, the baseline BioBERT classifier achieved a five-class accuracy of 47.60% and a 4-colour mapping accuracy of 49.95%, with a median inference latency of approximately 94 ms. To address severe class imbalance and study the safety trade-off between critical-case sensitivity and overall precision, stratified oversampling (text augmentation plus random minority oversampling) was applied during training. This raised Level 1 (Red) recall from 8.00% to 21.71% under external domain shift, though at a corresponding cost to overall accuracy. The system functions strictly as a rapid, nurse-overridable decision-support tool, expediting initial routing while preserving human clinical judgment and objective vital-sign scoring.

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ABBREVIATIONS

API Application Programming Interface

ASTS Adult South African Triage Scale

AVPU Alert, Voice, Pain, Unresponsive (consciousness scale)

BERT Bidirectional Encoder Representations from Transformers

BioBERT Biomedical Bidirectional Encoder Representations from Transformers

CE Cross-Entropy (classification loss)

CI Confidence Interval

CLS Classification token (`[CLS]`) used as the sequence summary in BERT

ED Emergency Department

ESI Emergency Severity Index

FFN Feed-Forward Network

FN False Negative (false clinical rejection for the gate; missed class in classification)

FP False Positive (false clinical acceptance for the gate; false alarm in classification)

GELU Gaussian Error Linear Unit

HIS Hospital Information System

HIT Hospital Information Technology

HR Heart Rate

JSON JavaScript Object Notation

KATH Komfo Anokye Teaching Hospital

KNUST Kwame Nkrumah University of Science and Technology

LMIC Low- and Middle-Income Country

MAE Mean Absolute Error (ordinal class distance)

MIMIC Medical Information Mart for Intensive Care

MIMIC-IV-ED MIMIC-IV Emergency Department module (v2.2 schema in this project)

MLM Masked Language Modelling

NHAMCS National Hospital Ambulatory Medical Care Survey

NLL Negative Log-Likelihood

NLP Natural Language Processing

OPD Outpatient Department

PAD Padding token used to fill sequences to length M

RR Respiratory Rate

SATS South African Triage Scale

SBP Systolic Blood Pressure

SEP Separator token (`[SEP]`) in BERT WordPiece sequences

SMOTE Synthetic Minority Over-sampling Technique

TEWS Triage Early Warning Score

TF-IDF Term Frequency–Inverse Document Frequency

TN True Negative

TP True Positive

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CHAPTER 1

INTRODUCTION

1.1 Background

Hospitals in Ghana operate under sustained pressure: crowded waiting areas, limited nursing staff, and long queues before a clinician sees a new arrival. Through triage, each arriving patient is assigned an urgency level so that scarce beds, nurses, and clinicians match clinical need rather than queue order alone. In well-resourced systems, urgency is reassessed as patients move through diagnostics, wards, and outpatient streams. In resource-constrained settings, however, standardised protocols such as the South African Triage Scale (SATS) are often applied only at the first desk. Once a patient leaves that point, their colour-coded acuity may no longer steer bed allocation, laboratory orders, or re-assessment timers. We call this failure *acuity discontinuity*.

Hospitals in Ghana, including KNUST University Hospital, apply SATS to structure intake across clinical departments, not only in a single emergency bay. Colour codes (Red, Orange, Yellow, Green, and Blue for dignity protocols) express how quickly a patient must be seen and where they should stream next: resuscitation, acute beds, urgent waiting, or routine outpatient care. Nurses at triage also compute a Triage Early Warning Score (TEWS) from objective vital signs. Yet many delays begin earlier, when a patient arrives with only words: a headache, fever, weakness, or abdominal pain described in free text before any thermometer or pulse oximeter reading is available.

This project proposes a digital *Flow Management Engine* in the form of a hospital chatbot for color-coded clinical pathways using Bidirectional Encoder Representations from Transformers (BERT). The chatbot accepts a single English free-text *chief complaint* (the primary reason the patient presents, in their own words; see Section 3.1.1) and returns an ESI-style acuity estimate, a SATS-aligned colour via a deterministic heuristic, and a clinical pathway card that can guide routing across outpatient departments, wards, and intake queues. The core contribution is mathematical: a pipeline from unstructured complaint text to acuity probabilities, colour, and pathway card, developed in Chapter 3. The deployed encoder is BioBERT, a biomedical BERT checkpoint fine-tuned for five-class triage. Crucially, this system is engineered strictly as a rapid, nurse-overridable decision-support tool. It expedites initial routing and queue management while strictly preserving the primacy of human clinical judgment and objective vital-sign scoring.

Hierarchy of evidence. Three layers must be kept distinct throughout this project. Empirically evaluated: BioBERT maps free text X to an ESI-style level $\hat{c} \in \{1, \ldots, 5\}$ (external metrics in Chapter 4). Proposed heuristic: $\hat{c}$ is converted to a SATS-aligned colour $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$ (Section 3.13); this is an operational crosswalk, not a claim of clinical equivalence between ESI and SATS. Protocol mapping: colour C_{NLP} indexes a pathway card $P(C)$ documented from SATS practice and an observational visit to KNUST University Hospital. Only the first layer is scored against gold acuity labels; the second and third are specified, auditable rules awaiting local clinical adjudication.

1.2 Motivation and significance

Three operational facts motivate the work.

1. **Text arrives before vitals.** At home, in a queue, or at a kiosk, patients can describe symptoms long before six TEWS parameters are measured. A language model that produces an initial SATS colour therefore fills a real temporal gap between arrival and nurse assessment (Stewart et al., 2023).

2. **Colour must travel with the patient.** SATS colours already exist in Ghanaian hospital policy (Rominski et al., 2014; KATH Emergency Department, 2018). The unsolved problem is not inventing colours, but binding them to destination, maximum waiting time $T_{\max}$ (the longest time allowed for that colour before the patient should be seen or reassessed), and escalation so that urgency does not vanish after the intake desk (South African Triage Group, 2012; Sundström et al., 2014).

3. **Green congestion harms Red and Orange care.** Non-urgent presentations occupying acute capacity (sometimes called the Green Tag Burden) delay time-critical patients (Yilmaz, 2021). Digital diversion of Green pathways toward outpatient or specialty streams is therefore a flow-management requirement, not a convenience feature.

For mathematics students and hospital stakeholders alike, the significance is twofold. First, the project shows how contemporary Transformer mathematics (tokenisation, embeddings, self-attention, Softmax) can be stated rigorously enough for examination while remaining deployable as a chatbot API. Second, it links that mathematics to an auditable protocol for healthcare facilities: every predicted acuity level maps through a fixed function f_{SATS} (which converts acuity levels 1–5 into a SATS colour) and then to a pathway card $P(C)$ (destination, $T_{\max}$, actions, and escalation for that colour).

1.3 Problem statement

Low- and middle-income country (LMIC) hospitals struggle with crowded waiting rooms and limited human resources at intake. Patients may wait hours before first assessment. Critical cases are not always identified quickly, while less urgent presentations congest acute zones. Medical systems are confusing; without immediate direction, patients do not know whether to join the emergency stream, the polyclinic, or self-care advice.

Manual, paper-based triage can create acuity discontinuity, rendering patient status static rather than dynamically responsive to clinical deterioration. This project organises the problem into five systemic issues, distinguishing what the literature documents, what an observational visit to KNUST University Hospital suggests, and what this project hypothesizes as addressable by a text-first Flow Management Engine:

1. **Pre-Assessment Bottleneck.** The literature identifies capacity-constrained first contact as a source of delay before clinical evaluation in busy LMIC emergency and intake settings (Rominski et al., 2014; Mitchell et al., 2023). Observations at KNUST University Hospital suggest that queues form before a structured chief complaint is captured digitally. This project hypothesizes that a gated chatbot can collect complaint text earlier without replacing nurse assessment.

2. **Acuity Degradation.** The literature identifies that urgency assigned at triage may not travel with the patient through imaging, laboratories, or wards when documentation is static (South African Triage Group, 2012; Sundström et al., 2014). This project hypothesizes that binding colour to destination, $T_{\max}$, and escalation on a pathway card can reduce loss of priority after the intake desk; countdown enforcement across departments was not measured prospectively here.

3. **Low-Acuity Congestion.** The literature identifies non-urgent (Green) load on acute capacity as harmful to time-critical care (Yilmaz, 2021). Observations at KNUST University Hospital suggest outpatient and specialty streams already exist for lower urgency. This project hypothesizes that automated Green diversion recommendations can support, not replace, local routing policy.

4. **Vital-Sign Latency.** The literature and SATS practice establish that TEWS requires measured vitals (South African Triage Group, 2012). At arrival, objective measurements are often incomplete, so high-risk language in the complaint is unused by TEWS until vitals are recorded. This project hypothesizes that an ESI-style text estimate can supply a preliminary colour for routing assist before TEWS is available, while TEWS remains the nurse vital-sign standard.

5. **Protocol Misalignment.** The literature on triage NLP often stops at label prediction without facility pathway cards (Stewart et al., 2023). Observations at KNUST University Hospital map colours to concrete departments. This project hypothesizes that a deterministic f_{SATS} plus $P(C)$ can make text predictions operationally actionable, subject to clinical validation of the ESI-to-SATS heuristic.

Without a framework to manage flow hospital-wide, overcrowding and resource mismatch persist; the most vulnerable patients can become invisible to the system after the first desk.

1.4 Aim and objectives

1.4.1 General objective

To design and specify a hospital chatbot that maps an unstructured English chief complaint to a SATS colour-coded clinical pathway, using a fine-tuned BioBERT classifier and protocol tables aligned to Ghanaian hospital intake.

1.4.2 Specific objectives

1. **Build a BioBERT classifier.** Predict an ESI-style urgency level (1–5) from a single unstructured English chief complaint using a fine-tuned BioBERT Softmax classification head.

2. **Map ESI-style acuity to SATS-aligned guidance.** Develop a deterministic mapping from predicted ESI-style acuity to a SATS-aligned colour and pathway recommendation for text-first intake assistance before vitals are taken, with TEWS kept as the parallel nurse vital-sign standard.

3. **Add a clinical gate.** Reject non-clinical chat so that non-medical messages receive no colour before BioBERT runs.

1.5 Research questions

Transitioning the theoretical capabilities of Transformer language models into a practical clinical tool requires systematic validation. To evaluate computational accuracy, operational specification, and safety posture of the proposed Flow Management Engine, this research seeks to answer the following questions.

1. Can a fine-tuned BioBERT model accurately classify unstructured English chief complaints into the correct ESI-style urgency level (1–5) under external domain shift?

2. How can a deterministic mapping heuristic (f_{SATS}) be used to transform predicted ESI-style acuity into SATS-aligned colour and pathway guidance before vital signs are available?

3. Can a mathematical clinical relevance gate detect and reject non-medical inputs before executing the Transformer classification model, as measured by accuracy, sensitivity, and specificity on a labeled clinical versus non-clinical contrast set?

4. Does training with additional Level 1 (Red) examples, to offset class imbalance, improve critical-case detection on the evaluation datasets, and at what cost to overall accuracy?

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

A comprehensive evaluation of published literature provides the necessary context for this project, encompassing hospital overcrowding, SATS and TEWS protocols, clinical pathways, Transformer language models, and natural language processing (NLP) in triage settings. Organized thematically, the subsequent review brings together current knowledge by allowing each section to build logically upon the last. This structure is deliberately designed to highlight the practical implications for deploying a triage chatbot within the Ghanaian healthcare system. Concluding the review, an explicit research gap is identified, establishing the theoretical justification for the text-based computational pipeline addressed in Chapter 3.

2.2 Hospital overcrowding and triage delay

Overcrowding in low- and middle-income country (LMIC) hospitals affects outpatient departments, emergency intake, and inpatient wards simultaneously (KATH Emergency Department, 2018; Mitchell et al., 2023). Delayed triage lengthens time-to-provider, increases left-without-being-seen rates, and correlates with poor outcomes for time-sensitive conditions. Waiting time interacts with colour assignment: when reassessment is weak, deterioration after the first desk can go unnoticed (Nurchis et al., 2025).

A structural driver is competition for acute capacity by non-urgent patients: the Green Tag Burden (Yilmaz, 2021). In LMICs, understaffing and limited diagnostics intensify that competition. Digital diversion of Green patients to outpatient, minors, or specialty streams is therefore a flow-management requirement for colour-coded pathways, not merely a convenience feature (Sundström et al., 2014).

Implication for this project. Any chatbot that assigns SATS colours must also attach destinations and timers; otherwise colour remains a label that dies at the desk (acuity discontinuity in Chapter 1). The next section reviews how SATS already supplies those colours and waiting-time rules in Ghanaian practice.

2.3 Ghanaian clinical context and SATS implementation

The South African Triage Scale (SATS) assigns four urgency colours for living patients (Red, Orange, Yellow, Green), each with a maximum waiting time $T_{\max}$ (the longest time allowed for each colour before the patient should be seen or reassessed) and destination rules (South African Triage Group, 2012). Blue is not a fifth urgency level: it marks deceased-on-arrival or dignity protocol. Implementation at Komfo Anokye Teaching Hospital (KATH) and related Ghanaian settings demonstrated that standardised colour coding can structure flow even when staffing is constrained, provided urgency information travels with the patient (Rominski et al., 2014; KATH Emergency Department, 2018).

SATS combines physiological early warning (TEWS) with clinical discriminators that can escalate colour when high-risk symptoms are present even if vitals are only mildly abnormal (for example central chest pain). Colour codes therefore already exist in hospital policy. The research gap is not inventing colours, but (i) assigning them from free-text complaints before full vitals, and (ii) binding colours to pathway protocols so acuity does not disappear after intake.

Hospitals such as KNUST University Hospital apply similar colour logic across clinical departments, not only in a single emergency bay. That multi-department setting motivates pathway cards that name outpatient urgent streams, surgery, obstetrics, pediatrics, and support units rather than a single undifferentiated "ED" destination. An observational site visit and staff inquiries informed the pathway tables in Chapter 3; those tables are the authors' documented interpretation of observed workflow and published SATS rules, not an official hospital protocol unless formally approved.

Local labeled triage NLP corpora for Ghanaian patient phrasing are not available at the scale of Triagegeist or NHAMCS. Formal English training and US external tests therefore leave cross-linguistic shift (Pidgin, Twi-influenced English) and local case-mix as open validation challenges rather than afterthoughts.

Colour alone still needs an objective vital-sign companion when measurements exist. The next section therefore turns to TEWS, the nurse-facing score that runs in parallel with any text-based colour.

2.4 TEWS and vital-sign triage

The Triage Early Warning Score (TEWS) is the deterministic vital-sign score used at SATS desks (South African Triage Group, 2012). For adult patients, nurses record seven parameters: mobility, respiratory rate (RR), heart rate (HR), systolic blood pressure (SBP), temperature, Alert–Voice–Pain–Unresponsive (AVPU) consciousness, and a trauma flag. Each parameter is converted into a small integer of points

using the SATS lookup tables (for example, very high heart rate, hypotension, or unresponsive consciousness contributes more points than a normal reading). Those seven contributions are then added to form a single total score T.

That integer T maps to a colour band: Red for $T \geq 7$, Orange for $5 \leq T \leq 6$, Yellow for $3 \leq T \leq 4$, and Green for $T \leq 2$. Because the map is fixed and auditable, two nurses who measure the same vitals obtain the same T and the same colour. TEWS is therefore objective; it does not read language. The formal sum and piecewise maps f_k are derived in Section 3.3.

Subjective symptoms such as "worst headache of my life" or "feverish and weak" may appear in the complaint long before corresponding vitals exist. TEWS cannot score a patient who has not yet been measured. Chapter 3 therefore presents TEWS mathematics as the hospital standard *parallel* to the text model: when vitals are available, nurses compute T; when they are not, the BioBERT path supplies an initial colour for routing assist. This project does not claim that Softmax replaces TEWS.

Colour and TEWS still need an operational destination. The next section reviews clinical pathways as the roadmaps that turn colour into where the patient should go and how soon.

2.5 Clinical pathways as operational roadmaps

Clinical pathways are step-by-step care roadmaps: destination, diagnostics, time targets, and escalation (Sundström et al., 2014). In SATS terms, colour without a pathway is incomplete: Red must mean immediate resuscitation access; Orange must mean a short countdown; Green must mean diversion away from acute beds when clinically appropriate (South African Triage Group, 2012; Yilmaz, 2021).

Pathway literature supports treating colour as operational, not decorative. Under-triage (assigning too low an urgency) is a recognised quality trigger in emergency medicine (Berkowitz et al., 2019); over-triage wastes scarce acute capacity. A chatbot that emits pathway cards $P(C)$ (for colour C: destination, $T_{\max}$, actions, and escalation) must therefore be evaluated not only on label accuracy but on whether routing rules are protocol-faithful and nurse-overridable.

To assign colour and pathway from free text before vitals, the project needs a language model that understands clinical wording. The next section reviews the Transformer mathematics that makes that possible.

2.6 Attention, BERT, and BioBERT

Sequential recurrent language models read text predominantly in one direction, limiting context for ambiguous clinical phrases. Vaswani et al. (2017) introduced scaled dot-

product self-attention, allowing each token to attend to all others in parallel within a fixed window. That property matters for triage language, where negation and severity modifiers ("no pain", "severe headache") must reshape neighbouring words.

Devlin et al. (2019) built BERT (Bidirectional Encoder Representations from Transformers) on this idea: masked language modelling (MLM) pre-training yields contextual embeddings

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(s)$$

where t_i is token i, E_{word}, E_{pos}, and E_{seg} are word, position, and segment embeddings, and s is the segment id (defined fully in Chapter 3). BERT is an encoder stack: it produces contextual vectors for classification and tagging rather than open-ended generation.

Lee et al. (2020) continued pre-training on PubMed and PubMed Central text to produce Biomedical BERT (BioBERT), a domain-specific encoder whose pre-training adapts the learned representations to biomedical language. Fine-tuning BioBERT on synthetic Triagegeist chief-complaint / ESI-style label pairs yields a supervised classifier from text to acuity levels. Classification with a fixed five-class label set reduces open-ended generative hallucination risk compared with unconstrained large language models that invent diagnoses in free prose (Stewart et al., 2023). The fixed map f_{SATS} then converts the predicted acuity level into a SATS-aligned colour for pathway routing.

Those models must still be judged against published triage NLP evidence. The next section reviews what that literature already shows and where it stops short for Ghanaian intake.

2.7 NLP for emergency and hospital triage

Recent evaluations of natural language processing in emergency triage highlight chief-complaint classification and clinical decision support as highly active, yet unevenly validated, research domains. For instance, Gligorijević et al. (2018) demonstrated the efficacy of deep attention models in stratifying emergency department patients based purely on unstructured text.

However, a critical vulnerability across these diagnostic models is severe class imbalance. Rare, life-threatening presentations (Level 1 / Red) are chronically under-represented compared to routine Yellow and Green cases. As a result, optimizing these models for global accuracy alone can obscure clinically unsafe under-triage, necessitating specialized statistical strategies to reduce the imbalance (Berkowitz et al., 2019; Chawla et al., 2002).

Furthermore, the majority of published NLP architectures are trained on high-income clinical corpora or electronic health records authored by medical professionals, rather

than raw patient phrasing. Most open-access triage benchmarks, such as MIMIC-IV-ED (Johnson et al., 2023a) and NHAMCS (National Center for Health Statistics (NCHS), 2024), are labelled using the five-level Emergency Severity Index (ESI) predominantly used in North America. The literature lacks a formalized framework to computationally bridge these resource-based ESI predictions to the physiological, colour-coded SATS pathways required for Ghanaian hospital intake, and it likewise lacks a mathematical pipeline that maps unstructured patient free text directly to SATS colours while binding those predictions to local departmental routing structures. While synthetic benchmarks such as Triagegeist (Laitinen-Fredriksson Foundation, 2025) and US-based datasets enable large-scale supervised fine-tuning, they introduce geographic and demographic domain shift. They remain a foundational computational step and do not circumvent the need for rigorous, prospective local validation in Ghanaian facilities.

Accuracy on labels is not the whole product. The next section reviews conversational triage tools and why this project's chatbot is deliberately constrained to colour and pathway cards.

2.8 Conversational agents and digital triage tools

Symptom-assessment applications and ED prediction chatbots show that conversational NLP can support triage-like advice before the nurse desk (Stewart et al., 2023). Validation studies emphasise safety, under-triage monitoring, nurse override, and comparison with conventional triage. Consumer tools often return free-form advice that is not mapped to hospital colour protocols or local destinations.

Our chatbot is deliberately constrained: it outputs acuity probabilities, a SATS colour, and a pathway card $P(C)$ aligned to SATS and Ghanaian facility routing, not free-form diagnosis essays. A clinical relevance gate rejects non-medical chat so that BioBERT is not run on greetings or sports conversation. That constrained design follows the safety emphasis in the triage NLP review literature.

2.9 Summary: what prior work supplies and what it leaves open

Table 2.1 summarises the strands reviewed above and shows how each strand feeds the next.

Strand	What exists	Open for this project
Overcrowding / Green burden	Evidence that delay and non-urgent congestion harm flow (Mitchell et al., 2023; Yilmaz, 2021)	Digital colour + diversion rules at Ghanaian intake
SATS / TEWS	Colour bands and vital-score algebra (South African Triage Group, 2012; Rominski et al., 2014)	Text-first colour before vitals; TEWS kept parallel
Pathways	Operational roadmaps and diversion models (Sundström et al., 2014)	Explicit $P(C)$ tables for seventeen departments
Transformers / BioBERT	Contextual encoders for biomedical text (Vaswani et al., 2017; Devlin et al., 2019; Lee et al., 2020)	Transparent pipeline $X \to \hat{y} \to C \to P(C)$
Triage NLP	ED text classifiers and reviews (Stewart et al., 2023; Gligorijević et al., 2018)	SATS-aligned, pathway-bound chatbot for LMIC intake

Table 2.1: Literature summary: available foundations versus the gap this project fills.

Table 2.2 situates representative triage NLP studies against this project on dataset, input, model, output, external validation, and Ghana/SATS pathway binding.

Study	Dataset	Input	Model	Output	Ext. val.	Ghana/SATS
Gligorijevic et al. (2018)	ED text	Free text	Deep attention	Acuity	Limited	No
Stewart et al. (2023) review	Multiple	Text ± EHR	Various NLP	Labels / CDS	Mixed	No
MIMIC-IV-ED / NHAMCS corpora	US ED	Complaint text	(eval only here)	ESI-style	Yes (US)	No
This project	Triagegeist train; MIMIC/ NHAMCS test	Chief complaint	BioBERT	ESI $\to f_{\text{SATS}}$ $\to P(C)$	Yes (US)	KNUST pathways

Table 2.2: Comparison of triage NLP settings. Ext. val. = evaluation outside the training corpus. Ghana/SATS = SATS-aligned pathway cards for a Ghanaian facility.

2.10 Research gap

Existing work separates vital-sign scores (TEWS), manual SATS colour assignment, and NLP classifiers trained largely abroad. In the literature reviewed for this project, we did not identify a unified mathematical account of how a free-text English chief complaint at Ghanaian hospital intake becomes a SATS-aligned colour and pathway card using

BioBERT, while TEWS remains documented as the nurse-facing vital standard and nurses retain override.

In short, prior art provides colours, scores, and growing NLP evidence; the reviewed sources do not provide the integrated text path

$$X \xrightarrow{\tau} H^{(0)} \xrightarrow{\text{BioBERT}} \hat{\mathbf{y}} \xrightarrow{f_{\text{SATS}}} C_{\text{NLP}} \rightarrow P(C_{\text{NLP}})$$

where X is the chief complaint, τ is tokenisation, $H^{(0)}$ is the input embedding matrix, $\hat{\mathbf{y}}$ is the ESI-style acuity distribution, f_{SATS} is the proposed heuristic from acuity level to SATS-aligned colour C_{NLP}, and $P(C_{\text{NLP}})$ is the pathway card (destination, $T_{\max}$, actions, escalation), all developed in Chapter 3, with clinical gating, Softmax framed as a Categorical distribution over mutually exclusive acuity levels, and department routing tables matched to Ghanaian healthcare facilities. The ESI-to-SATS step remains a proposed heuristic (Section 3.13), distinct from empirically scored ESI-style classification.

This project addresses three specific gaps in the current triage literature:

1. **Transparent Mathematics.** Many NLP triage papers report final accuracy without showing the underlying mathematical steps. This project explicitly details every transformation, from tokenisation to the classification Softmax, in a consistent notation.

2. **Actionable Pathways.** Most models output a bare urgency label and stop. This project binds that label to a SATS-aligned pathway, providing specific waiting times, escalation rules, and department destinations mapped to a Ghanaian hospital setting.

3. **Parallel TEWS Integration.** Some literature treats text classifiers as a complete replacement for vital signs. This project explicitly positions the Transformer text-path as an antecedent routing assist, retaining TEWS as the independent, deterministic standard for objective vital-sign scoring.

Chapter 3 fills that gap for the text path; Chapter 4 reports evaluation.

CHAPTER 3

METHODOLOGY

Translating unstructured chief complaints into actionable South African Triage Scale (SATS) classifications requires a rigorous computational pipeline. To bridge the underlying mathematical architecture with its practical application, the following sections employ a recurring illustrative example modeled on a standard patient presentation at KNUST University Hospital: *"I have a headache and feel feverish. My body aches and I feel weak."* This end-to-end transformation applies to all textual inputs that successfully clear the clinical relevance gate detailed in Section 3.15. Crucially, this architecture operates strictly as a clinical decision-support mechanism. It provides probabilistic routing guidance to assist triage staff, rather than functioning as an autonomous diagnostic tool or replacing definitive clinical judgment.

3.1 Methodological overview

The hospital chatbot draws on two complementary ideas from clinical practice; this project develops the text-based path in full and documents the Triage Early Warning Score (TEWS) as the parallel vital-sign standard.
The chapter addresses the three specific objectives from Chapter 1 in order:

1. Build a BioBERT Softmax classifier that turns one English chief complaint into urgency probabilities over levels 1–5. Tokenisation, embeddings, the encoder, Softmax, and Stage 3 fine-tuning are derived in the sections that follow (especially Sections 3.9 and 3.12).

2. Deploy colour and pathway guidance from text at intake before vitals are taken. After the Softmax chooses a level, the fixed map f_{SATS} names the SATS colour and the pathway card $P(C)$ states destination, maximum waiting time $T_{\max}$, actions, and escalation, while TEWS remains the nurse vital-sign path in parallel (Section 3.3).

3. Add a clinical relevance gate so non-clinical chat receives no colour. Before BioBERT runs, the gate scores how clinical the message is and withholds colour from greetings and other non-medical text.

First, **clinical pathways** encode hospital protocols: symptoms and context map to an urgency colour and a destination (resuscitation bay, acute bed, urgent waiting,

outpatient stream). Pathways are the product the chatbot shows to patients and staff after classification.

Second, TEWS is the nurse-facing vital-sign score used at SATS triage desks (Adult TEWS fully in Section 3.3). The chatbot's primary engine for *free-text chief complaints* at intake is BioBERT, not a weighted sum of vitals.

When vitals are not yet available, the text model below supplies the intake colour. Nurses may later record vitals and apply TEWS independently; this project specifies the natural language processing (NLP) path only.

3.1.1 Chief complaints

The *chief complaint* is the primary reason the patient presents, expressed in their own words (or an attendant's) as unstructured free text at intake, before vitals, physical examination, or formal diagnosis. Nurses and triage staff traditionally record this as prose on paper or electronic forms; it captures symptom onset, severity, and context (for example, "headache and feverish, body aches, feel weak"). It is not a diagnosis, not a TEWS vital-sign score, and not a full medical history: it is only the presenting problem text X that this project classifies.

The chatbot accepts one **English** chief-complaint string per submission. The running example *"I have a headache and feel feverish. My body aches and I feel weak."* illustrates a typical Yellow-level presentation used throughout this chapter; pipeline symbol X appears in Table 3.1 and Figure 3.1.

3.1.2 Training data provenance

Model training and external evaluation draw on distinct data layers.

Training corpus (not published test). Stage 3 fine-tuning uses the Triagegeist synthetic emergency-department benchmark released by the Laitinen-Fredriksson Foundation on Kaggle (Laitinen-Fredriksson Foundation, 2025). Structured intake records (`train.csv`, acuity labels) are inner-joined with free-text rows (`chief_complaints.csv`) on `patient_id`, yielding $N = 80{,}000$ pairs of English free-text chief complaints with synthetic ESI-style urgency levels 1–5. Vitals and demographics in the source files are excluded so the NLP path matches single-shot text input at intake (Section 3.12.3). Labels map to SATS colours only via the proposed heuristic f_{SATS} (Section 3.13); the corpus is **not** SATS-labelled. This corpus was subjected to a stratified 90/10 split (seed 42), resulting in approximately 72,000 pairs for training and 8,000 pairs for internal validation. The 72,000-pair training set was used exclusively to update the model weights during backpropagation. The 8,000-pair validation set was isolated strictly to monitor validation loss and select the optimal training checkpoint. Crucially, neither of these Triagegeist subsets was used to evaluate

the model's final clinical performance. The evaluation protocol was designed to reduce the risk of data leakage and synthetic memorisation: all published accuracy, colour accuracy, and under/over-triage metrics in Chapter 4 are computed on separate external cohorts (the MIMIC-IV-ED open demo matching the v2.2 triage schema (Johnson et al., 2023b,a) and NHAMCS 2018–22 (National Center for Health Statistics (NCHS), 2024)), not on Triagegeist holdout text. This corpus was **not** collected at KNUST University Hospital, and the model has not yet been clinically tested on real patients at KNUST University Hospital.

KNUST-informed contextual examples. Following an observational site visit to KNUST University Hospital, the authors mapped existing clinical pathways and department structures through direct observation of intake workflows and stakeholder inquiries with hospital staff. Based on these established protocols, the authors generated supplementary English chief complaints to validate pathway routing and illustrate the pipeline (for example, *"I have a headache and feel feverish. My body aches and I feel weak."* in Table 3.24); this supports qualitative analysis only. They are **not** merged into the 80,000-pair fine-tuning corpus.

3.1.3 BERT and BioBERT (preview)

Older language models read text left-to-right only. BERT uses bidirectional self-attention and masked language modelling so each token depends on left and right context (Devlin et al., 2019; Vaswani et al., 2017). BioBERT continues that pre-training on biomedical text (Lee et al., 2020). Architecture constants (layers, hidden size, heads) and Stage 1 MLM are derived in Sections 3.9 and 3.12.1; Stage 3 fine-tuning on Triagegeist is in Section 3.12.

3.1.4 The `[CLS]` summary token

Classification does not read all token outputs separately. An artificial token `[CLS]` is inserted at position $i = 0$. After the encoder stack, its hidden state $h_{[\mathrm{CLS}]}$ summarises the entire complaint and feeds the Softmax head (Section 3.6); layer count and dimension are fixed in Section 3.9.

3.1.5 Pipeline summary

Figure 3.1 previews the end-to-end natural language processing (NLP) path from a free-text chief complaint to a SATS colour and pathway card. Each box corresponds to one transformation; the subsections and sections that follow derive the formulas, symbols, and worked steps for every stage.

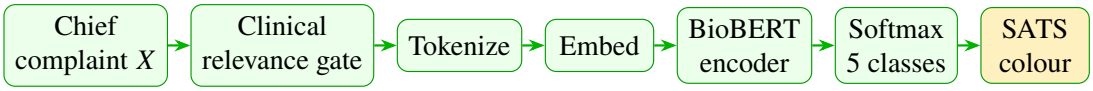


Figure 3.1: NLP pipeline from chief complaint to SATS colour (TEWS is documented separately in Section 3.3).

1. **Chief complaint X.** The patient or attendant submits one unstructured English text string describing symptoms and relevant history. That string is the only clinical input to the NLP path at this stage: no vital signs, no diagnosis, and no full medical history are required before the model can propose a colour.

2. **Clinical relevance gate.** A rule-based score $g(X)$ decides whether the message is clinical enough to classify. Greetings and other non-medical chat are rejected here so BioBERT does not invent an urgency colour for text that is not a chief complaint.

3. **Tokenize.** WordPiece splits X into subword tokens and inserts the special markers `[CLS]`, `[SEP]`, and `<PAD>`. The result is a fixed-length sequence of integer IDs that the embedding table can look up.

4. **Embed.** Each token ID becomes a vector in the BioBERT embedding space (Section 3.7); positional and segment terms are added to form $H^{(0)}$.

5. **BioBERT encoder.** The stacked encoder applies self-attention across tokens (Section 3.9).

6. **Softmax.** The `[CLS]` summary feeds a five-class Softmax head (Section 3.11); the mode is the predicted ESI-style level.

7. **SATS colour and pathway.** The predicted level $\hat{c}$ maps to colour $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$, and the pathway card $P(C_{\text{NLP}})$ names destination, $T_{\max}$, actions, and escalation.

Each stage above is derived and exemplified later in this chapter. The end-to-end summary restates the full path for the implemented prototype.

3.1.6 Safety and nurse authority

The Flow Management Engine is engineered strictly as a clinical decision-support system; it is structurally constrained from acting as an autonomous diagnostic or discharge authority. To preserve the primacy of human clinical judgment, the architecture enforces three strict operational boundaries throughout the pipeline:

1. **Algorithmic Gating.** A mathematical relevance gate actively filters non-clinical text; non-medical inputs are systematically rejected prior to BioBERT classification.

2. **Probabilistic Output Constraint.** The classification head outputs a Categorical distribution $\hat{\mathbf{y}}$ rather than a binary safety certificate. Low-confidence classifications and near-ties explicitly mandate immediate escalation to human triage staff.

3. **Preservation of Objective Scoring.** When physical vitals are acquired, TEWS and SATS discriminators govern bedside clinical colour. The NLP text-path operates as an antecedent routing assist and is programmed to never overwrite objective physiological data.

Pathway cards therefore propose destinations and timers that nurses can accept, modify, or reject. That principle is repeated in the results discussion (Section 4.5) and when Softmax confidence is low.
Table 3.1 records the mathematical object entering and leaving each stage so the reader can follow how one complaint string becomes a colour and pathway card.

Stage	**Input**	**Output**
Chief complaint	Patient text string	X (raw string)
Clinical gate	X	Pass/fail via $g(X)$
Tokenize	X	Token sequence $(t_1, \ldots, t_M)$, $M \leq 128$
Embed	Token IDs	Input matrix $H^{(0)} \in \mathbb{R}^{M \times 768}$
BioBERT	$H^{(0)}$	Final matrix $H^{(12)}$; summary $h_{[\mathrm{CLS}]}$
Softmax	$h_{[\mathrm{CLS}]}$	Acuity probabilities $\hat{\mathbf{y}} \in \mathbb{R}^5$
SATS colour	$\hat{\mathbf{y}}$	Level $\hat{c}$, colour C_{NLP}, pathway $P(C_{\mathrm{NLP}})$

Table 3.1: Pipeline stage inputs and outputs (NLP path).

3.2 SATS colour chart

SATS assigns each living patient one of four urgency colours (Red, Orange, Yellow, Green) with a maximum waiting time $T_{\max}$ and hospital destination (South African Triage Group, 2012). Figure 3.2 shows that living urgency scale separately from Blue. Blue marks deceased-on-arrival or dignity protocol; it is *not* a fifth urgency level and is never produced by the Softmax head.

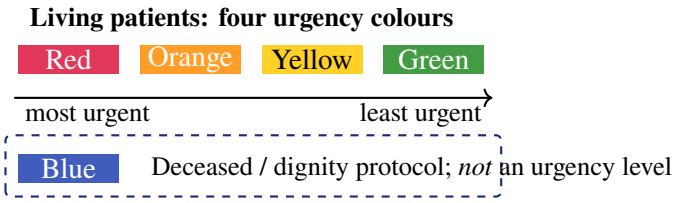


Figure 3.2: SATS living urgency colours (Red–Green) and Blue dignity protocol (not an urgency band).

Colour	**Meaning**	$T_{\max}$	**Typical destination**
Red	Immediate life threat	0 min	Resuscitation bay
Orange	Very urgent	10 min	Acute / high-dependency bed
Yellow	Urgent	60 min	Emergency waiting / urgent stream
Green	Non-urgent	< 4 h	Outpatient department (OPD) / minors / polyclinic
Blue	Dignity protocol	N/A	Mortuary / private space

Table 3.2: SATS colours, waiting targets, and routing (SATS-aligned).

Remark 3.1. *Red–Green are the four living urgency colours used for routing assist. Blue lies outside that scale (dignity / deceased protocol) and is never emitted by the chatbot. The Softmax path outputs one of Red, Orange, Yellow, or Green via f_{SATS} so intake staff can route living patients without re-interpreting free text.*

For the illustrative complaint in this chapter, the NLP model in Section 3.11 may assign moderate urgency (Yellow) with high probability; the same framework applies regardless of presenting symptoms.

3.3 TEWS: Triage Early Warning Score

TEWS is the deterministic vital-sign score used at SATS triage desks (South African Triage Group, 2012). The SATS Training Manual publishes three age-appropriate charts: younger child, older child, and adult. This project documents only **Adult TEWS** (Figure 14 of that manual): patients **older than 12 years** or **taller than** 150 **cm**. The adult bands are recorded in this thesis as Table 3.4. Paediatric TEWS (Figures 12–13 of the manual) use different RR/HR bands and different mobility labels ("normal for age" rather than Walking / With help / Stretcher/immobile); those charts are out of scope here because the chatbot has no age or height feature.

3.3.1 Definition

Let $\mathbf{v} = (v_1, \ldots, v_7)$ denote adult heart rate (HR), respiratory rate (RR), systolic blood pressure (SBP), mobility, temperature, AVPU consciousness, and trauma flag. Each parameter maps to points $f_k(v_k) \in \{0, 1, 2, 3\}$ via the Adult TEWS lookup in Table 3.4. TEWS aggregates these contributions into one integer so that nurses at different desks can compare adult patients on a common scale. Weights $w_k = 1$:

$$T = \sum_{k=1}^{7} w_k \, f_k(v_k) = \sum_{k=1}^{7} f_k(v_k). \tag{3.1}$$

Summing seven bounded terms yields $T \in \{0, \ldots, 17\}$: the adult maxima are Mobility 2, RR 3, HR 3, SBP 3, Temp 2, AVPU 3, and Trauma 1 (Table 3.4). Typical triage totals are much lower. Because each adult f_k follows that table, the score is reproducible and auditable: two nurses who measure the same adult vitals obtain the same T.

k	Parameter v_k	Source	Unit / scale
1	Heart rate (HR)	SATS Adult TEWS	beats/min
2	Respiratory rate (RR)	SATS Adult TEWS	breaths/min
3	Systolic blood pressure (SBP)	SATS Adult TEWS	mmHg
4	Mobility	SATS Adult TEWS	Walking / With help / Stretcher/immobile
5	Temperature	SATS Adult TEWS	°C
6	AVPU consciousness	SATS Adult TEWS	alert / voice / pain / unresponsive (+ confused)
7	Trauma	SATS Adult TEWS	present / absent (any injury within 48 h)

Table 3.3: Adult TEWS parameters used in Table 3.4.

Remark 3.2. *Each adult vital sign contributes zero to three points. Elevated HR, abnormal RR, low or very high SBP, or reduced consciousness increase T; normal adult values contribute zero.*

3.3.2 Piecewise scoring functions (adult only)

Adult f_k bands below match Table 3.4 (eligibility in the section open). Each f_k returns an integer in $\{0, 1, 2, 3\}$.

Adult heart rate f_1 (beats/min):

$$f_1(v_1) = \begin{cases} 3 & v_1 > 129 \\ 2 & v_1 < 41 \text{ or } 111 \le v_1 \le 129 \\ 1 & 41 \le v_1 \le 50 \text{ or } 101 \le v_1 \le 110 \\ 0 & 51 \le v_1 \le 100 \end{cases} \tag{3.2}$$

Adult respiratory rate f_2 (breaths/min):

$$f_2(v_2) = \begin{cases} 3 & v_2 > 29 \\ 2 & v_2 < 9 \text{ or } 21 \le v_2 \le 29 \\ 1 & 15 \le v_2 \le 20 \\ 0 & 9 \le v_2 \le 14 \end{cases} \tag{3.3}$$

Adult systolic blood pressure f_3 (mmHg):

$$f_3(v_3) = \begin{cases} 3 & v_3 < 71 \\ 2 & 71 \le v_3 \le 80 \text{ or } v_3 > 199 \\ 1 & 81 \le v_3 \le 100 \\ 0 & 101 \le v_3 \le 199 \end{cases} \tag{3.4}$$

Adult mobility f_4, temperature f_5, AVPU f_6, and trauma f_7 (adult mobility labels Walking / With help / Stretcher/immobile):

$$f_4(v_4) = \begin{cases} 0 & \text{walking} \\ 1 & \text{with help} \\ 2 & \text{stretcher / immobile} \end{cases}, \tag{3.5}$$

$$f_5(v_5) = \begin{cases} 2 & v_5 < 35 \text{ or cold, or } v_5 > 38.4 \text{ or hot} \\ 0 & 35.0 \le v_5 \le 38.4 \end{cases}, \tag{3.6}$$

$$f_6(v_6) = \begin{cases} 3 & \text{unresponsive} \\ 2 & \text{confused or reacts to pain} \\ 1 & \text{reacts to voice} \\ 0 & \text{alert} \end{cases}, \tag{3.7}$$

$$f_7(v_7) = \begin{cases} 1 & \text{trauma present (any injury within 48 h)} \\ 0 & \text{no trauma} \end{cases}. \tag{3.8}$$

Parameter	3	2	1	0	1	2	3
Mobility				Walking	With help	Stretcher/immobile	
RR		< 9		9–14	15–20	21–29	> 29
HR		< 41	41–50	51–100	101–110	111–129	> 129
SBP	< 71	71–80	81–100	101–199		> 199	
Temp		Cold or < 35		35–38.4		Hot or > 38.4	
AVPU		Confused		Alert	Voice	Pain	Unresponsive
Trauma				No	Yes		

Table 3.4: Adult TEWS lookup, adapted from Figure 14 of the SATS Training Manual (South African Triage Group, 2012).

3.3.3 Colour map

The mapping $C_{\text{TEWS}}(T)$ converts the integer score into the same colour vocabulary used at the SATS desk, so vital-sign triage aligns with hospital routing rules (South African Triage Group, 2012):

$$C_{\text{TEWS}}(T) = \begin{cases} \text{Red} & T \geq 7 \\ \text{Orange} & 5 \leq T \leq 6 \\ \text{Yellow} & 3 \leq T \leq 4 \\ \text{Green} & T \leq 2 \end{cases} \tag{3.9}$$

where T is the TEWS total from Equation (3.1) and $C_{\text{TEWS}}(T)$ is the vital-sign colour band. Thresholds follow the standard SATS implementation (TEWS 7 or more is Red): a single point difference can change the colour band, so nurses record each f_k carefully. The diagram below visualises these bands on the score axis.

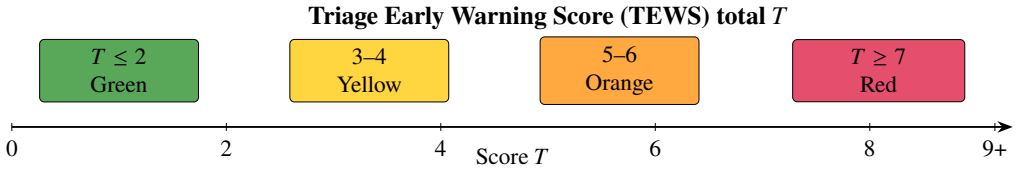


Figure 3.3: TEWS score ranges mapped to SATS colours (segments separated for readability).

3.3.4 Relationship to the text Softmax path

Both paths use the four living SATS urgency colours, but TEWS is the adult seven-feature sum $T = \sum_k f_k(v_k)$ and colour map above, while Softmax is a Categorical distribution over five acuity levels from complaint text then f_{SATS} (Section 3.9). They are complementary: text may be the only early signal; after nurse assessment, TEWS

and clinical discriminators remain authoritative. Formally

$$X \xrightarrow{\text{BioBERT}} C_{\text{NLP}}, \qquad V \xrightarrow{\text{TEWS}} C_{\text{TEWS}}.$$

Disagreement $C_{\text{NLP}} \neq C_{\text{TEWS}}$ is a review signal, not proof that the text model is wrong. This project documents both, implements Softmax fully, and never treats TEWS absence as a chatbot limitation: nurses already compute TEWS when vitals exist.

Aspect	**TEWS**	**BioBERT Softmax**
Input	Seven vitals / flags v_k (incl. SBP)	English chief complaint X
Core map	Integer sum T	Probabilities $\hat{\mathbf{y}}$
Colour rule	$C_{\text{TEWS}}(T)$ bands	$f_{\text{SATS}}(\hat{c})$
When available	After measurement	At text submission
Authority	Nurse desk standard	Intake assist (overridable)

Table 3.5: Side-by-side comparison of TEWS and the text Softmax path.

k	Parameter	Measured v_k	$f_k(v_k)$	Running sum
1	HR	125 bpm	2	2
2	RR	26 breaths/min	2	4
3	SBP	120 mmHg	0	4
4	Mobility	walking	0	4
5	Temperature	36.8 °C	0	4
6	AVPU	alert	0	4
7	Trauma	absent	0	4
			Total T:	**4**

Table 3.6: Term-by-term *adult* TEWS calculation for HR = 125 bpm, RR = 26 breaths/min, SBP = 120 mmHg, all other adult parameters scoring zero on Table 3.4.

Example 3.1. *Suppose an adult patient has vitals HR* = 125 *bpm, RR* = 26 *breaths/min, and SBP* = 120 *mmHg; mobility walking, temperature* 36.8 °C*, AVPU alert, and trauma absent. Table 3.6 lists each adult f_k contribution against Table 3.4; summing gives*

$$\begin{aligned}
f_1(125) &= 2 && \textit{(adult HR band } 111\text{–}129\textit{)},\\
f_2(26) &= 2 && \textit{(adult RR band } 21\text{–}29\textit{)},\\
f_3(120) &= 0 && \textit{(adult SBP band } 101\text{–}199\textit{)},\\
f_k(\cdot) &= 0 && k = 4, \ldots, 7,\\
T &= 2 + 2 = 4,\\
C_{\text{TEWS}}(4) &= \textit{Yellow} \quad \textit{(Equation } (3.9)\textit{)}.
\end{aligned}$$

Every term matches Table 3.4: RR = 26 scores 2 on the adult 21–29 band. If only these vitals were measured, the score is partial but still auditable. At intake, before vitals exist, the BioBERT path in Section 3.11 supplies the text-based colour.

3.4 Neural networks, Transformers, and BioBERT

The text path is a neural network: stacked maps from token vectors to acuity probabilities. Motivation for BERT/BioBERT is in Section 3.1.3; layer width, attention, and Stage 1 MLM are in Sections 3.9 and 3.12.1.
The input representation preview (full detail in Section 3.7) is

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(s), \tag{3.10}$$

where the three terms are added element-wise (not concatenated). Table 3.7 summarises each term for a single chief complaint ($s = 0$).

Term	**Meaning**	**Example in triage**
E_{word}	Token identity: which word or subword?	"headache" vs "feverish"
$E_{\text{pos}}(i)$	Position in the sequence: order matters	"head ache" ≠ "ache head"
$E_{\text{seg}}(s)$	Sentence segment s (A or B in paired inputs)	Always $s = 0$ for one complaint

Table 3.7: Breakdown of the three terms in Equation (3.10).

The remainder of this chapter specifies English input X, the clinical gate, WordPiece tokenisation τ, embeddings, the encoder, Softmax over ESI-style levels, colour map f_{SATS}, and pathway card $P(C)$.

3.5 Tokenisation

3.5.1 Input X

The NLP path begins with a single English *chief complaint* string X: the primary reason the patient presents, in free text. It is not a diagnosis, not a TEWS vital-sign vector, and not a full medical history. Non-clinical messages may be rejected by the relevance gate (Section 3.15) before tokenisation runs. Only clinical X proceeds into the encoder.

3.5.2 The tokenisation map τ

Neural networks require fixed-size numerical input. Tokenisation is the first transformation after the gate: a map τ sends raw complaint text X into a bounded sequence of subword tokens with integer IDs.

$$X \xrightarrow{\tau} (t_1, \ldots, t_M), \quad M \leq 128. \tag{3.11}$$

Here X is the English chief-complaint string, τ is the **WordPiece tokenisation map** used by BERT and BioBERT, $t_1, \ldots, t_M$ are the resulting subword tokens (with special markers), and M is the sequence length capped at 128. The map τ is a *deterministic discrete encoding* that segments the string using a fixed vocabulary and emits token IDs. It is **not** a neural network layer; it does not learn during fine-tuning. Learning begins at the embedding lookup that follows.

3.5.3 Why the maximum length is $M = 128$

Native BioBERT / BERT-base supports sequences up to length 512. This project **deliberately uses** maximum sequence length $M = 128$ tokens for Stage 3 fine-tuning and chatbot inference. The shorter cap has three roles:

- **Batching.** Padding every complaint to length 128 yields rectangular tensors so many requests can share the same matrix multiply at intake.
- **Bound on context.** At most 128 subword positions (including `[CLS]`, `[SEP]`, and pads) enter one forward pass; longer free text must be truncated before submission.
- **Latency.** Shorter sequences reduce attention cost relative to the native 512 limit, which matters for sub-second intake response.

Remark 3.3. *The fixed length $M = 128$ is a* project deployment choice*, not a claim that BioBERT cannot handle longer inputs: the architecture default remains 512, while this chatbot truncates and pads to 128 for batching and latency.*

WordPiece keeps common clinical words (“fever”, “pain”) whole and splits rare or complex forms (e.g. “headache” $\rightarrow$ `head+##ache`). Vocabulary $\mathcal{V}$ has $|\mathcal{V}| \approx 28{,}996$ entries; each token ID indexes a row in the embedding table. When a word splits into two subwords, it occupies **two positions** in the sequence (see rows 4 and 7 in Table 3.9), so M counts subword tokens, not raw words.

How the WordPiece algorithm works in detail. BioBERT ships with a fixed vocabulary of character and subword pieces learned offline; Stage 3 fine-tuning does not rebuild that vocabulary. At inference, each whitespace-separated word in X is segmented by a left-to-right greedy longest-match search: starting at the first character, the algorithm

takes the longest vocabulary entry that matches a prefix of the remaining string, emits that piece, then continues on the rest of the word. Continuation pieces after the first are marked with the prefix ## so the model knows they belong to the same original word (for example head then ##ache). If no vocabulary entry matches, the algorithm falls back to smaller pieces, ultimately to individual characters when necessary. Each emitted piece is then replaced by its integer ID in $\mathcal{V}$. Common clinical words such as "fever" or "pain" usually remain one token; rarer compounds such as "headache" or "feverish" split, which keeps the vocabulary manageable while still representing informal chief-complaint spelling. Those IDs are exactly the indices looked up in the embedding matrix in the next stage.

Three special tokens structure every input sequence:

- [CLS] (ID 101) is inserted at position $i = 0$ as the classification summary token; its final hidden state feeds the acuity head (Section 3.6).
- [SEP] (ID 102) is the **separator** token marking the end of the complaint sentence. It tells the model where clinical text stops, so subsequent <PAD> fillers are not interpreted as symptoms.
- <PAD> (ID 0) fills unused positions up to length $M = 128$; the attention mask sets these positions to zero so they do not affect encoding.

Token	**ID**	**Purpose**
[CLS]	101	Classification summary; always first
[SEP]	102	End-of-sentence boundary before padding
<PAD>	0	Batch padding to fixed length 128

Table 3.8: Special tokens in the BioBERT tokenizer.

i	Raw	Token	ID
0	n/a	[CLS]	101
1	I	i	1045
2	have	have	2031
3	a	a	1037
4	headache	head+##ache	7994, 8772
5	and	and	1998
6	feel	feel	2514
7	feverish	fever+##ish	9643, 6804
8	.	.	1012
9	n/a	[SEP]	102
10–127	n/a	<PAD>	0

Table 3.9: Token IDs for the illustrative complaint (padding to length 128 omitted).

3.6 The [CLS] and [SEP] tokens

Sequence classification requires one fixed-length vector summarising the entire complaint. Rather than averaging all token rows (which would weight filler words equally with symptoms), BERT introduces an artificial summary token [CLS] at position $i = 0$. The [SEP] token at the end of the real text marks the sentence boundary: encoding stops treating tokens after [SEP] as part of the clinical complaint, which matters once <PAD> fillers occupy the remaining positions. Only the [CLS] row is passed to the classification head; [SEP] itself is not used for acuity prediction.
After $L = 12$ encoder layers:

$$h_{[\mathrm{CLS}]} = H^{(L)}_{0,:} \in \mathbb{R}^{768}, \tag{3.12}$$

where $H^{(L)}$ is the final hidden-state matrix after all twelve layers and $H^{(L)}_{0,:}$ denotes its first row (the [CLS] position).

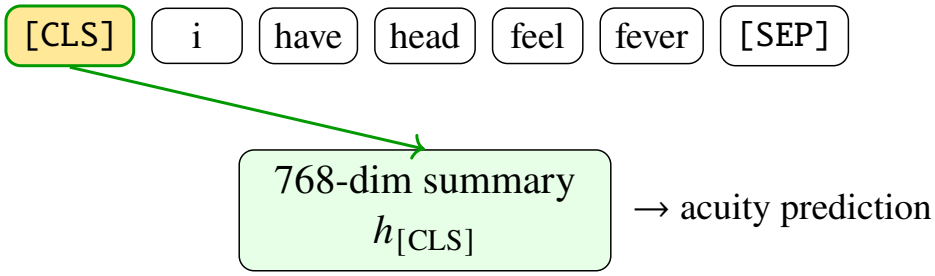


Figure 3.4: [CLS] aggregates context into one summary vector for classification.

Self-attention updates the [CLS] row at every layer so that it encodes information from all symptom tokens; the classification head therefore reads a single vector that represents the full clinical wording of X.

3.7 Embedding lookup

A token ID is an index, not a meaning. The embedding layer maps each ID to a dense vector in $\mathbb{R}^{768}$ via the learned matrix $E_{\mathrm{word}} \in \mathbb{R}^{V \times 768}$:

$$\mathbf{e}_i = E_{\mathrm{word}}[\mathrm{id}(t_i)] \in \mathbb{R}^{768}, \tag{3.13}$$

where t_i is the token at position i and $\mathrm{id}(t_i)$ is its integer vocabulary index (the WordPiece ID in Tables 3.8–3.9).

Symbol	Definition
$V = \lvert\mathcal{V}\rvert$	Vocabulary size ($\approx$ 28,996)
$\mathrm{id}(t_i)$	Vocabulary index of token t_i
$D = 768$	Hidden dimension (BioBERT base configuration)
$\mathbf{e}_i$	Word embedding vector for token at position i

Table 3.10: Embedding lookup notation.

The lookup $\mathbf{e}_i$ is one term in the full input representation $E(t_i) = E_{\text{word}} + E_{\text{pos}} + E_{\text{seg}}$ from Equation (3.10). After PubMed pre-training, BioBERT adapts these representations to biomedical language.

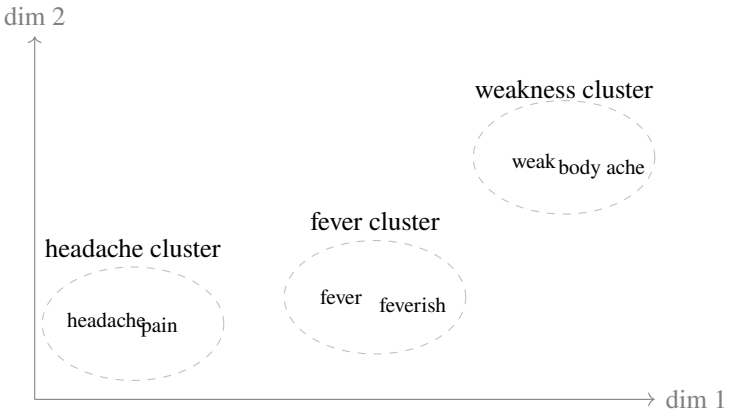


Figure 3.5: Schematic of BioBERT embedding space: familiar symptom words group together after biomedical pre-training (two examples per cluster).

When a patient writes “feverish” instead of “fever”, or “head ache” as two tokens, the model can still map the wording into regions of this space associated with similar presentations. That generalisation across varied patient language is central to text-first triage.

Contextual mapping sums three embeddings (Equation (3.10)). The positional embedding depends only on index i, and the segment embedding depends on segment id s (always $s = 0$ for a single complaint):

$$E_{\text{pos}}(i) \in \mathbb{R}^{768}, \qquad E_{\text{seg}}(s) \in \mathbb{R}^{768}, \quad s = 0. \tag{3.14}$$

The final input vector for token t_i at position i is the element-wise sum across all $D = 768$ dimensions:

$$E(t_i)_d = E_{\text{word}}(t_i)_d + E_{\text{pos}}(i)_d + E_{\text{seg}}(s)_d, \quad d = 1, \ldots, 768. \tag{3.15}$$

In words: for each dimension d, the word number, the position number, and the segment number are added together. The result is still a single 768-length vector for that token,

not three vectors glued end to end.

3.7.1 Why three embedding components

The model does not use a single lookup. Each token must pass through **three** additive components before the encoder stack, for distinct reasons:

1. **Word embedding E_{word}.** Encodes *what* the subword is (clinical meaning of "feverish" versus "football").

2. **Position embedding $E_{\text{pos}}(i)$.** Encodes *where* the token sits in the complaint. Transformers have no built-in left-to-right recurrence; without position, "severe pain then mild" could not be distinguished from the reverse order.

3. **Segment embedding $E_{\text{seg}}(s)$.** Marks sentence A versus B in paired BERT tasks. For a single chief complaint we always use $s = 0$; the term is kept so the BioBERT checkpoint remains architecturally compatible even though a second segment is unused at intake.

Omitting word embeddings would remove meaning; omitting position would scramble order-sensitive clinical phrasing; omitting segment would break compatibility with the pre-trained BERT/BioBERT weight layout. The three components are therefore not decorative: they are the minimal input interface the Transformer expects.

3.7.2 Why the hidden dimension is $D = 768$

BioBERT-base uses hidden width $D = 768$. That width is fixed by the published architecture (Lee et al., 2020; Devlin et al., 2019) and matches

$$12 \text{ attention heads} \times 64 \text{ dimensions per head} = 768. \tag{3.16}$$

Every token vector, every encoder layer, and the classification summary $h_{[\text{CLS}]}$ live in $\mathbb{R}^{768}$. The width is large enough to encode rich biomedical semantics yet small enough to stack twelve layers and train on large corpora. Changing D would require a different model family; this project therefore keeps $D = 768$ throughout.

Symbol	Definition
$E_{\text{word}}(t_i)$	Row of E_{word} looked up by token ID
$E_{\text{pos}}(i)$	Learned vector for position i in the sequence
$E_{\text{seg}}(s)$	Learned vector for segment s (here $s = 0$)
$E(t_i)_d$	Component d of the summed embedding

Table 3.11: Symbols in the full input embedding (Equation (3.15)).

Example 3.2. *For token "head" at position $i = 4$, suppose the first three dimensions (illustrative only) are*

$$\begin{aligned} E_{\text{word}} &= [0.31,\ -0.18,\ 0.52], \\ E_{\text{pos}}(4) &= [0.05,\ 0.02,\ -0.01], \\ E_{\text{seg}}(0) &= [0.00,\ 0.00,\ 0.00], \\ E(t_4) &= E_{\text{word}} + E_{\text{pos}}(4) + E_{\text{seg}}(0) \\ &= [0.36,\ -0.16,\ 0.51] \quad \textit{(first three dimensions)}. \end{aligned}$$

The same element-wise addition repeats for all remaining dimensions $d = 4, \ldots, 768$. The result is row 4 of $H^{(0)}$.

Remark 3.4. *The embedding table is like a dictionary: each token ID points to a list of 768 numbers encoding meaning. Adding position tells the model where the word appears; adding segment (zero here) reserves the mechanism for two-sentence inputs in other BERT tasks.*

Figure 3.6 shows the flow for one token position ($D = 768$; why this width is fixed is in Section 3.7).

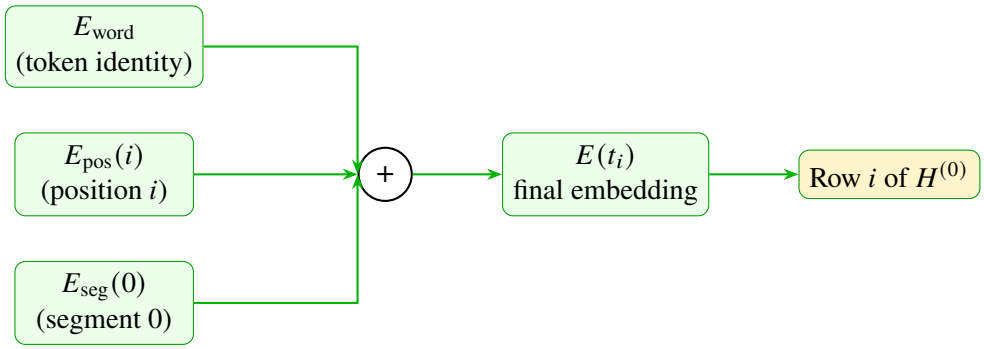


Figure 3.6: Three embeddings summed element-wise for one token position.

3.8 Input matrix $H^{(0)}$

Stacking the embedding vector for each of the M tokens forms the initial hidden-state matrix

$$H^{(0)} \in \mathbb{R}^{M \times 768}, \tag{3.17}$$

where $M \leq 128$ is the padded sequence length and 768 is the BioBERT hidden dimension. **Rows versus columns.** By convention in this project:

- **Rows** ($M \leq 128$) index **token positions** in the sequence (row 0 is `[CLS]`, then the WordPiece tokens, then `[SEP]` and pads).

- **Columns** (768) index **hidden feature dimensions** of the BioBERT representation.

Row i is built by applying Equation (3.15) at position i: look up the word vector, add the positional vector $E_{\text{pos}}(i)$, add the segment vector $E_{\text{seg}}(0)$, and stack the result. Viewed this way, a chief complaint is a table of numbers that the encoder layers progressively transform. The same row/column convention holds for $H^{(\ell)}$ after each encoder layer: shape stays $M \times 768$; only the entries change as self-attention mixes information *across rows*.

Later, the classification weight matrix $W \in \mathbb{R}^{5\times 768}$ uses the complementary layout: **rows** are the five acuity classes and **columns** are the 768 features that multiply $h_{[\text{CLS}]}$ (Section 3.11).

i	**Token**	D_1	D_2	D_3	D_4	$\cdots$
0	`[CLS]`	0.02	0.11	−0.05	0.33	$\cdots$
1	i	−0.08	0.04	0.19	0.01	$\cdots$
2	have	0.15	−0.22	0.07	0.44	$\cdots$
4	head	0.31	−0.18	0.52	0.09	$\cdots$
7	fever	0.44	0.27	−0.11	0.61	$\cdots$
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Table 3.12: Example submatrix of $H^{(0)}$ (first four of 768 dimensions; illustrative values).

After encoding, row 0 (`[CLS]`) holds the summary passed to the softmax classifier; the remaining rows are not used directly for the final colour decision.

3.9 Encoder layers

BioBERT applies the same encoder block $L = 12$ times. Each block first lets tokens attend to one another (mixing symptom phrases), then applies a feed-forward network (FFN) to each row independently. **Residual connections** add the block input back to its output so that gradients can flow through depth during training. **Layer normalisation** (LayerNorm) rescales each row to zero mean and unit variance for numerical stability (Vaswani et al., 2017; Devlin et al., 2019). MultiHeadAttention is the twelve-head self-attention block of Section 3.10. GELU is the Gaussian Error Linear Unit used in BERT/BioBERT FFNs (smooth nonlinear activation; not ReLU). The update equations are

$$Z^{(\ell)} = \text{LayerNorm}\big(H^{(\ell-1)} + \text{MultiHeadAttention}(H^{(\ell-1)})\big), \tag{3.18}$$

$$H^{(\ell)} = \text{LayerNorm}\big(Z^{(\ell)} + \text{FFN}(Z^{(\ell)})\big), \tag{3.19}$$

$$\text{FFN}(x) = W_2\,\text{GELU}(W_1 x + b_1) + b_2, \tag{3.20}$$

where $H^{(\ell-1)}$ and $H^{(\ell)}$ are the hidden-state matrices before and after layer ℓ, $Z^{(\ell)}$ is the intermediate state after attention, and FFN uses $W_1 \in \mathbb{R}^{768\times3072}$, $W_2 \in \mathbb{R}^{3072\times768}$, and biases $b_1 \in \mathbb{R}^{3072}$, $b_2 \in \mathbb{R}^{768}$ in the standard BERT-base configuration. The inner dimension 3072 is four times the hidden width 768, giving the FFN enough capacity to transform each token row nonlinearly. In plain terms, each layer updates every token using the whole complaint, while the residual connection keeps the previous representation so meaning is refined rather than replaced.

Component	**What it does**
MultiHeadAttention	Twelve parallel self-attention heads; each learns different word relationships
FFN	Two linear layers with GELU: expands each row to 3072 dims, then projects back to 768
LayerNorm	Rescales activations for stable training across depth
Residual (+)	Adds sub-layer input to output so earlier information is not lost

Table 3.13: Components inside one BioBERT encoder layer.

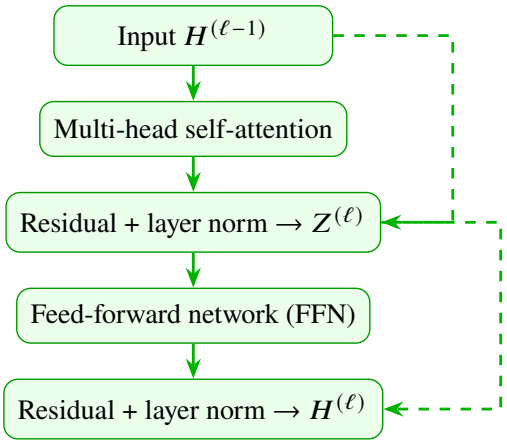


Figure 3.7: One BioBERT encoder layer (dashed arrows: residual connections). The block is stacked $L = 12$ times.

In the attention sub-step, each token row is updated using information from every other row in the complaint, which is how "headache" and "feverish" influence one another before acuity is estimated. The FFN sub-step then applies the same nonlinear transformation to each row separately, increasing representational capacity. Repeating this block twelve times transforms $H^{(0)}$ into $H^{(12)}$:

$$H^{(0)} \xrightarrow{\text{Layer 1}} H^{(1)} \xrightarrow{\text{Layer 2}} \cdots \xrightarrow{\text{Layer 12}} H^{(12)} = H^{(L)}.$$

Shallow layers (1–4) tend to encode individual words and short phrases; middle layers (5–8) combine symptoms such as "headache" with "feverish" and "weak"; deep layers

(9–12) encode broader urgency patterns. All M token rows are processed in parallel at each layer, not left-to-right as in older language models.

Remark 3.5. *Stacking layers is like reading a complaint once for words, once for symptom combinations, and once for overall urgency. The* `[CLS]` *row at layer 12 becomes* $h_{[\mathrm{CLS}]}$ *for classification.*

3.10 Self-attention

Self-attention computes a weighted combination of token representations for each position. The computation proceeds in four steps, matching Figure 3.8:

1. **Project** hidden rows into query, key, and value matrices.
2. **Score** pairwise compatibility between tokens.
3. **Normalise** scores into attention weights.
4. **Combine** value rows into context-aware outputs.

Let $H = H^{(\ell-1)} \in \mathbb{R}^{M\times 768}$ be the input to the current encoder layer. Learned projection matrices $W_Q, W_K, W_V \in \mathbb{R}^{768\times d_k}$ with key dimension $d_k = 64$ produce

$$\begin{aligned} Q = HW_Q, \quad K = HW_K, \quad V = HW_V, \\ S = QK^\top/\sqrt{d_k}, \\ A = \mathrm{softmax}(S), \\ \mathrm{output} = AV, \end{aligned}$$

where $Q, K, V \in \mathbb{R}^{M\times d_k}$, S and A are $M \times M$ score and weight matrices, and softmax is applied row-wise so each query distributes weights that sum to 1.
Following Vaswani et al. (2017), one attention head uses:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{3.21}$$

with key dimension $d_k = 64$ per head. For one query row, the softmax over keys assigns weight α_j to token j:

$$\alpha_j = \frac{\exp(e_j)}{\sum_{k=1}^{M}\exp(e_k)}, \quad \sum_{j=1}^{M}\alpha_j = 1, \tag{3.22}$$

where e_j is the scaled score in row j of $QK^\top/\sqrt{d_k}$ and M is the sequence length (at most 128). BioBERT runs twelve such heads in parallel; their outputs are concatenated and projected back to 768 dimensions so each head can specialise (e.g. symptom co-occurrence vs negation).

Remark 3.6. *In triage language: Q is what the current token looks for in the rest of the complaint; K is the label each other token shows; V is the meaning that is mixed in when the match is strong. Dividing by $\sqrt{d_k}$ keeps the match scores from becoming too large before Softmax. The Softmax inside Equation* (3.21) *only mixes* ***token*** *meanings; it is not the five-class Softmax that later chooses the colour.*

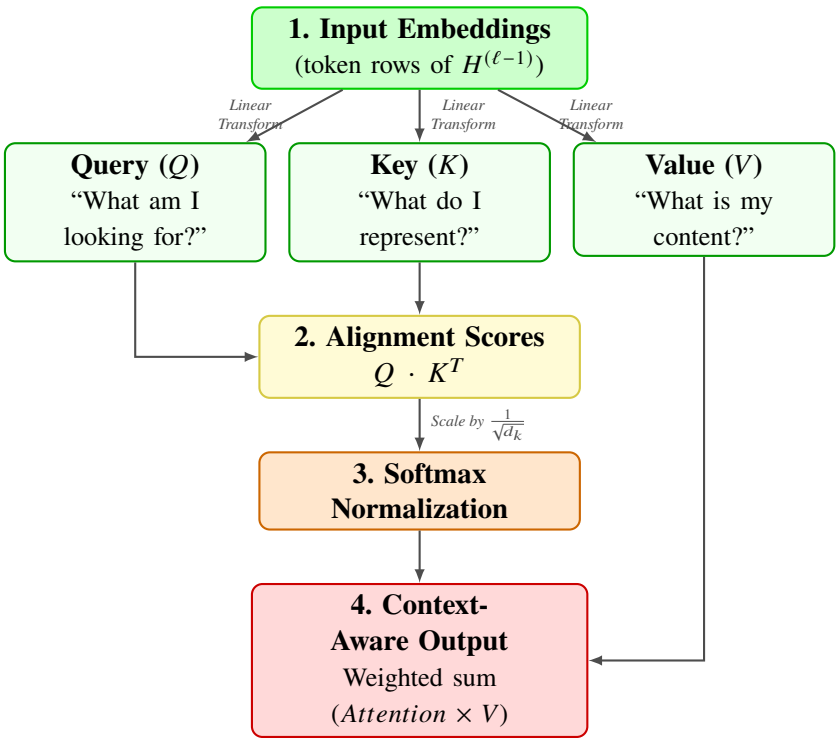


Figure 3.8: Self-attention data flow within one encoder layer (schematic architecture diagram; not extracted model weights).

Symbol	Definition
$H = H^{(\ell-1)}$	Hidden-state matrix entering the current encoder layer ($M \times 768$)
W_Q, W_K, W_V	Learned projection matrices ($768 \times d_k$) that form queries, keys, and values
Q, K, V	Query, key, and value matrices ($M \times d_k$) derived from H
$QK^\top$	Matrix of pairwise compatibility scores between tokens
$\sqrt{d_k}$	Scaling constant ($d_k = 64$) that prevents dot products from growing too large
e_j	Raw attention score for token j before normalisation
α_j	Normalised attention weight for token j (a probability)
M	Sequence length (number of tokens, at most 128)

Table 3.14: Self-attention notation.

Query token	[CLS]	headache	feverish	filler
Attention to	0.09	0.32	0.41	0.18

Table 3.15: Schematic attention weights when the query is [CLS] (illustrative, not measured from the model). Symptom tokens receive higher weight than filler words.

The product $QK^\top$ measures how strongly each token should attend to every other token. Dividing by $\sqrt{d_k}$ keeps those scores in a range where the softmax in Equation (3.22) does not saturate. In a complaint mentioning both "headache" and "feverish", attention typically assigns higher weight to symptom tokens than to function words such as "I" or "and", so the updated representation emphasises clinically informative wording (Table 3.15).

3.11 Classification softmax

3.11.1 Mutually exclusive acuity categories

When a patient submits a chief complaint, their clinical urgency for routing is treated as falling into **exactly one** of five ESI-style acuity levels $\{1, 2, 3, 4, 5\}$. A presentation cannot be Level 1 (immediate) and Level 5 (routine) at the same decision instant. Because only one level is selected for the pathway colour, the five categories are modelled as **mutually exclusive**. That modelling choice determines the output layer: the network must produce a distribution over five options, not five independent yes/no scores.
If the head used five independent sigmoid units, each class probability could be high at once (for example $\hat{y}_1 = 0.9$ and $\hat{y}_5 = 0.8$), which would not define a single routing colour. Softmax avoids that inconsistency by normalising across the five logits together.

3.11.2 Softmax as the probability translator

After BioBERT encoding, the classification head maps the summary vector $h_{[\mathrm{CLS}]} \in \mathbb{R}^{768}$ to five unconstrained real scores called **logits**:

$$\begin{aligned} \mathbf{z} &= W h_{[\mathrm{CLS}]} + \mathbf{b}, \\ \hat{\mathbf{y}} &= \mathrm{softmax}(\mathbf{z}), \\ \hat{y}_c &= \frac{\exp(z_c)}{\sum_{j=1}^{5} \exp(z_j)}, \end{aligned} \tag{3.23}$$

with $W \in \mathbb{R}^{5\times 768}$ (**rows** = acuity classes, **columns** = features), $\mathbf{b} \in \mathbb{R}^5$, logits z_c, and $\sum_{c=1}^{5} \hat{y}_c = 1$. Softmax enforces the two elementary rules of a finite probability distribution: every $\hat{y}_c$ is non-negative, and the five values sum to one. The predicted level is $\hat{c} = \arg\max_c \hat{y}_c$. Expressing output as probabilities allows downstream routing

logic to consider confidence (for example a dominant Yellow at about 69% versus a near tie between Yellow and Green).

Remark 3.7. *The vector* $\mathbf{z}$ *holds five* ***raw scores (logits)****, not probabilities: a larger* z_c *only means that urgency level looks stronger. Softmax turns those scores into five chances that add to 1. The largest chance is the acuity level used for the SATS colour.*

This **classification Softmax** (over five acuity classes) is distinct from the **attention Softmax** inside each encoder layer (over token positions). Both use the same functional form; they act on different axes of the model. Attention Softmax asks "which tokens matter for this query?"; classification Softmax asks "which acuity level is most probable for this complaint?"

Remark 3.8. *The linear map* $\mathbf{z} = Wh_{[\mathrm{CLS}]} + \mathbf{b}$ *is the triage analogue of a multi-class "score sheet": each row of* W *emphasises features in* $h_{[\mathrm{CLS}]}$ *that favour one acuity level. Softmax then converts those raw scores into a comparable probability vector for hospital routing.*

3.11.3 Categorical distribution ($K = 5$)

By producing five probabilities that sum to one, Softmax parameterises a **Categorical distribution** over the acuity levels. A Categorical distribution is the discrete law for exactly K mutually exclusive outcomes; here $K = 5$. The vector $\hat{\mathbf{y}}$ is the parameter of that law:

$$\hat{y}_c = P(\text{acuity} = c \mid X), \qquad c \in \{1, 2, 3, 4, 5\}, \tag{3.24}$$

where X is the chief complaint and $\hat{y}_c$ is the Softmax probability of acuity level c. Informally, $\hat{\mathbf{y}}$ behaves like a weighted five-sided die whose face probabilities are determined by the complaint text. Sampling from that die would draw a random acuity; hospital routing instead takes the mode $\hat{c} = \arg\max_c \hat{y}_c$ (and may escalate to a nurse when the mode is not dominant).

Stage 3 training maximises the probability of the ESI-style label under this Categorical model. Equivalently, it minimises the negative log-likelihood (NLL)

$$\mathcal{L}_{\mathrm{NLL}} = -\log \hat{y}_y = -\log P(y \mid X), \tag{3.25}$$

where y is the true acuity label, X is the complaint, and $\hat{y}_y = P(y \mid X)$ is the Softmax probability of that label. This is exactly the cross-entropy loss used in Section 3.12 when labels are one-hot.

3.11.4 Bernoulli as the $K = 2$ building block

The **Bernoulli** distribution is the two-outcome special case ($K = 2$): a single trial with results $\{0, 1\}$ and success probability p. Softmax with two logits (z_0, z_1) yields

$$p = \frac{\exp(z_1)}{\exp(z_0) + \exp(z_1)} = \sigma(z_1 - z_0), \tag{3.26}$$

where z_0 and z_1 are the two class logits, p is the probability of class 1, and $\sigma(u) = 1/(1 + e^{-u})$ is the logistic sigmoid. That identity shows why binary logistic regression is Softmax with $K = 2$. The five-class triage head is the same mathematical idea generalised from a weighted coin to five mutually exclusive faces.

Example 3.3 (Bernoulli building block)**.** *Suppose a toy two-class head distinguishes "urgent" versus "non-urgent" with logits* $z_{\mathrm{U}} = 1.0$ *and* $z_{\mathrm{N}} = -0.5$*. Then*

$$\begin{aligned} \exp(z_{\mathrm{U}}) &= e^{1.0} \approx 2.72, \\ \exp(z_{\mathrm{N}}) &= e^{-0.5} \approx 0.61, \\ P(\text{urgent}) &= \frac{2.72}{2.72 + 0.61} \approx 0.82. \end{aligned}$$

Raising z_{U} *relative to* z_{N} *increases urgent probability exactly as raising one acuity logit relative to the others increases that class probability in the five-class Softmax.*

Symbol	Definition
$h_{[\mathrm{CLS}]}$	768-dimensional summary vector from final encoder layer
$W, \mathbf{b}$	Learned classification weight matrix (5×768) and bias
z_c	Logit (pre-softmax score) for acuity level c
$\hat{y}_c$	Categorical probability of level $c \in \{1, \ldots, 5\}$
$\hat{c}$	Predicted level: index of largest $\hat{y}_c$

Table 3.16: Classification head notation.

Example 3.4 (Five-class Softmax for the running complaint)**.** *Suppose illustrative logits*

for levels 1–5 are

$$\begin{aligned}
\mathbf{z} &= [-1.2,\ -0.5,\ 1.8,\ 0.3,\ -0.9],\\
\exp(\mathbf{z}) &\propto [0.30,\ 0.61,\ 6.05,\ 1.35,\ 0.41],\\
\sum_{c=1}^{5} \exp(z_c) &= 8.72,\\
\hat{\mathbf{y}} &\approx [0.03,\ 0.07,\ 0.69,\ 0.15,\ 0.05],\\
\hat{c} &= 3,\\
f_{\text{SATS}}(3) &= \textit{Yellow}.
\end{aligned}$$

Level 3 has the largest Categorical probability ($\hat{y}_3 \approx 0.69$, rounded to 72% in some deployment logs; Section 3.13). The mass on neighbouring Orange (0.07) and Green (0.15 + 0.05) shows residual uncertainty: Softmax does not claim certainty, only a ranked distribution suitable for pathway assist with nurse oversight when confidence is low.

Remark 3.9. *Softmax probabilities are calibrated only to the extent that the training distribution matches live intake. High $\hat{y}_c$ on Triagegeist validation text is not a clinical guarantee at KNUST University Hospital; low-confidence or contradictory presentations remain mandatory nurse review.*

3.12 Model training pipeline

The deployed triage classifier is not trained from random initialisation. It is the product of three sequential stages: biomedical pre-training (Stage 1), encoder transfer from a prior triage checkpoint (Stage 2), and project fine-tuning on synthetic Triagegeist chief-complaint / ESI-style label pairs (Stage 3). Stages 1 and 2 supply medical language and partial triage adaptation; Stage 3 is the comprehensive retraining that maps free-text complaints to five acuity levels for this chatbot.

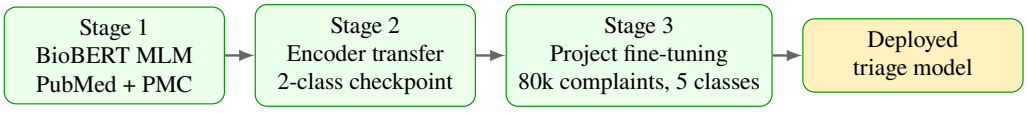


Figure 3.9: Three-stage training stack for the BioBERT triage classifier.

3.12.1 Stage 1: BioBERT pre-training (masked language modelling)

Stage 1 supplies the biomedical MLM prior from PubMed/PMC (Lee et al., 2020); the Hugging Face identifier is in Table 3.18. No triage labels are used. Random tokens are

masked and predicted from bidirectional context:

$$\mathcal{L}_{\text{LM}} = -\sum_{i \in \mathcal{M}} \log P(t_i \mid H^{(L)};\, \theta), \tag{3.27}$$

where $\mathcal{M}$ is the set of masked positions, t_i is the true token at i, $H^{(L)}$ is the final encoder state (Section 3.9), θ denotes model parameters, and $P(t_i \mid \cdot)$ is the probability of the correct word. Stage 1 does **not** teach triage urgency, SATS colours, or informal chief-complaint phrasing.

Symbol	Definition
$\mathcal{M}$	Set of masked token positions in the sentence
t_i	True (correct) word at masked position i
θ	All model weights and biases
$P(t_i \mid H^{(L)};\, \theta)$	Model probability for the correct token
$\mathcal{L}_{\text{LM}}$	Total MLM penalty; lower means better predictions

Table 3.17: Masked language modelling notation (Stage 1).

Example 3.5. *If the masked token should be "headache", the MLM contribution is*

Good prediction ($P = 0.92$):

$$-\log(0.92) \approx 0.08 \qquad \textit{(small penalty)},$$

Poor prediction ($P = 0.01$):

$$-\log(0.01) \approx 4.6 \qquad \textit{(large penalty)}.$$

3.12.2 Stage 2: Encoder transfer from a triage checkpoint

Stage 2 warm-starts from the intermediate triage checkpoint in Table 3.18 (Yuvrajxms09, 2024): a BioBERT encoder already partially adapted to urgency-related text with a **two-class** head ($W \in \mathbb{R}^{2\times 768}$). That 2-class head is **discarded** on load because our labels require five acuity levels. The loader uses `ignore_mismatched_sizes=True`: only encoder weights carry forward; the output layer is randomly reinitialised for five classes. Stage 2 therefore contributes triage-oriented encoder features without constraining the final colour mapping.

Stage	Hugging Face identifier	Role in this project
1	`dmis-lab/biobert-base-cased-v1.2`	BioBERT-Base v1.2 (DMIS Lab / Lee et al.); biomedical MLM prior (Lee et al., 2020; DMIS Lab, 2020)
2	`Yuvrajxms09/biobert-triage-classifier`	Intermediate 2-class triage checkpoint; encoder kept, head discarded (Yuvrajxms09, 2024)
3	Project fine-tune (local weights)	Five-class Softmax on Triagegeist; published NHAMCS/MIMIC metrics

Table 3.18: Exact model identifiers for reproducibility of Stages 1–3.

3.12.3 Stage 3: Fine-tuning on synthetic Triagegeist chief-complaint / ESI-style label pairs

Stage 3 is the project fine-tune on Triagegeist text–label pairs (corpus size, 90/10 split, and vitals exclusion in Section 3.1.2). A new classification head $W \in \mathbb{R}^{5\times768}$, $\mathbf{b} \in \mathbb{R}^5$ is trained from scratch while encoder weights update with a small learning rate.

Level	Count	Proportion	SATS colour
1 (most urgent)	3,222	0.0403	Red
2	13,439	0.1680	Orange
3	28,921	0.3615	Yellow
4	23,020	0.2878	Green
5 (least urgent)	11,398	0.1425	Green
Total	**80,000**	**1.000**	

Table 3.19: Class distribution in the 80,000 labeled chief-complaint training pairs.

Hyperparameter	Value
Epochs	3
Batch size (train and eval)	16
Learning rate α	2×10^{-5}
Weight decay	0.01
Max sequence length	128 tokens
Mixed precision (FP16)	Enabled
Total optimizer steps	13,500 (3 epochs $\times$ 4,500 steps)
Best validation loss	0.001812 (epoch 3)

Table 3.20: Fine-tuning hyperparameters (Stage 3).

Component	Updated during Stage 3?
Embedding layers	Yes (small learning-rate updates)
12 transformer encoder layers	Yes
Classification head $W, \mathbf{b}$	Yes (trained from scratch)

Table 3.21: Which model components change during project fine-tuning.

The objective is **cross-entropy loss**: for each training example, the model is penalised when it assigns low probability to the nurse's true acuity label. For a single example with true class y,

$$\mathcal{L}_{\text{CE}} = -\log \hat{y}_y, \tag{3.28}$$

where $y \in \{1, \ldots, 5\}$ is the ESI-style acuity label and $\hat{y}_y$ is the Softmax probability the model assigns to that correct level. Over the full training set the same symbol denotes the empirical risk

$$\mathcal{L}_{\text{CE}} = -\sum_{i=1}^{N} \log P(y_i \mid X_i;\, \theta), \quad N = 80{,}000, \tag{3.29}$$

where X_i is complaint i, y_i is its label, θ denotes all trainable parameters, and $P(y_i \mid X_i;\, \theta) = \hat{y}_{y_i}$ is the predicted probability of the true class.

Example 3.6. *If the nurse labeled a complaint Yellow (level 3), the cross-entropy penalty is*

Low confidence on the true class ($\hat{y}_3 = 0.05$):

$$\mathcal{L}_{\text{CE}} = -\log(0.05) \approx 3.0 \qquad \textit{(large penalty)},$$

High confidence on the true class ($\hat{y}_3 = 0.95$):

$$\mathcal{L}_{\text{CE}} = -\log(0.95) \approx 0.05 \qquad \textit{(small penalty)}.$$

Fine-tuning adjusts weights so correct classes receive higher probability on similar complaints.

Backpropagation computes how each weight contributed to $\mathcal{L}_{\text{CE}}$ by applying the chain rule backward from the loss through the softmax head and all twelve encoder layers. For classification weights $W \in \mathbb{R}^{5\times 768}$, with logits $\mathbf{z} = Wh_{[\text{CLS}]} + \mathbf{b}$ where $h_{[\text{CLS}]} \in \mathbb{R}^{768}$ is the final summary vector and $\mathbf{b} \in \mathbb{R}^5$ is the bias,

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial W} = \frac{\partial \mathcal{L}_{\text{CE}}}{\partial \hat{\mathbf{y}}} \cdot \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{z}} \cdot \frac{\partial \mathbf{z}}{\partial W}. \tag{3.30}$$

In words: (i) $\partial \mathcal{L}_{\text{CE}}/\partial \hat{\mathbf{y}}$ measures how wrong the predicted probabilities are; (ii) $\partial \hat{\mathbf{y}}/\partial \mathbf{z}$ propagates through softmax; (iii) $\partial \mathbf{z}/\partial W$ links logits to the weight matrix. The same

logic applies layer by layer through BioBERT.
Gradient descent then adjusts weights in the direction that reduces the loss:

$$W_{\text{new}} = W_{\text{old}} - \alpha \frac{\partial \mathcal{L}_{\text{CE}}}{\partial W}, \tag{3.31}$$

where $\alpha = 2 \times 10^{-5}$ is the learning rate, W_{old} is the current weight matrix, and W_{new} is the updated matrix. When the model confidently predicts the wrong colour, backpropagation increases the loss sharply and the subsequent weight update pushes similar complaints toward the ESI-style class on the next training pass.

Remark 3.10. *Pre-training (Stage 1) gives the model medical language; comprehensive fine-tuning (Stage 3) teaches triage decisions on short, informal chief-complaint phrasing. The learning rate is kept small so PubMed knowledge is preserved while the head and encoder adapt to urgency labels.*

3.12.4 Mitigating class imbalance

Table 3.19 shows that Level 3 dominates Triagegeist while Level 1 (Red) is scarce. The published *stratified-oversampling* checkpoint therefore rebalances the **training** split only (the validation split is left untouched) before a second Stage 3 run, with embedding-space SMOTE disabled (`USE_EMBEDDING_SMOTE=False`). The algorithmic steps are:

1. Restrict imbalance mitigation to the stratified Triagegeist training rows from Section 3.12.3.

2. Apply light **text augmentation** to minority urgent classes (acuity levels 1 and 2): synonym or paraphrase variants of the chief-complaint string, with protected medical keywords left unchanged. Configured yields are two augmented variants per Level 1 row and one per Level 2 row.

3. Apply **random minority oversampling** with replacement until target counts $n_1 =$ 10,000 and $n_2 =$ 15,000; Levels 3–5 keep their natural training counts.

4. Run end-to-end Stage 3 fine-tuning (WordPiece → BioBERT → Softmax) on the expanded training set with the same hyperparameters as Table 3.20.

Chapter 4 contrasts this checkpoint with the unrebalanced baseline and explains why classical embedding SMOTE was not chosen (Section 4.2.5).

3.13 Mapping acuity to colour: f_{SATS}

3.13.1 ESI-to-SATS mapping assumption

The Softmax head predicts Emergency Severity Index (ESI)–style acuity levels 1–5 learned from synthetic Triagegeist training pairs and scored on external MIMIC-IV-ED and NHAMCS labels. Those sources are **not** SATS-labelled: ESI is primarily resource-based, whereas SATS combines vital-sign TEWS with clinical discriminators. No direct SATS colour labels are used for training. Hospital routing at KNUST University Hospital nonetheless uses SATS colours, so this project defines a fixed deterministic operational heuristic f_{SATS} that converts an ESI-style level into a SATS-aligned colour for text-first routing assist:

$$f_{\text{SATS}}(\hat{c}) = C_{\text{NLP}} = \begin{cases} \text{Red} & \hat{c} = 1 \\ \text{Orange} & \hat{c} = 2 \\ \text{Yellow} & \hat{c} = 3 \\ \text{Green} & \hat{c} \in \{4, 5\} \end{cases} \tag{3.32}$$

where $\hat{c} = \arg\max_c \hat{y}_c$ is the predicted acuity level, f_{SATS} is the fixed level-to-colour map, and C_{NLP} is the resulting SATS-aligned colour from the text path.

ESI uses five ordered acuity levels. SATS, by contrast, has only four urgency colours for living patients (Red, Orange, Yellow, Green); Blue is reserved for deceased-on-arrival or dignity protocol and is not another urgency band. The heuristic f_{SATS} therefore maps the five ESI-style levels onto those four living colours (with Levels 4 and 5 both Green). That structural bridge does *not* imply clinical equivalence. The map is a proposed operational crosswalk for intake assistance, not a validated clinical translation, and requires adjudication against clinician-assigned SATS colours before operational claims beyond routing assist.

ESI-style level	**Clinical name**	$f_{\text{SATS}}(\ell)$
1	Resuscitation	Red
2	Emergent	Orange
3	Urgent	Yellow
4	Less urgent	Green
5	Non-urgent	Green

Table 3.22: Proposed ESI-style acuity → SATS colour heuristic for text-first routing (not a clinical equivalence table).

Levels 4 and 5 both map to Green because both are non-urgent in the ESI-style five-level schema, while SATS living urgency stops at four colours. Blue (dignity / deceased

protocol) is never emitted by this five-class head. Colour accuracy is therefore not equivalent to five-class accuracy: f_{SATS} is many-to-one on $\{4, 5\}$.

3.14 Clinical pathways $P(C_{\text{NLP}})$

Once C_{NLP} is known, the chatbot attaches a pathway card that specifies waiting time, destination, and escalation rules drawn from SATS-aligned hospital protocol at KNUST University Hospital:

$$P(C_{\text{NLP}}) = \big(T_{\max},\ \text{destination},\ \text{actions},\ \text{escalation}\big), \tag{3.33}$$

where $P(C_{\text{NLP}})$ is the pathway card for colour C_{NLP}, $T_{\max}$ is the maximum waiting time, destination is the primary clinical department, actions list support steps (nursing, pharmacy, diagnostics), and escalation defines what happens if the timer expires or the patient worsens. Colour alone is not enough at intake: the pathway card names where the patient should go and how soon they should be seen. The four tuple components correspond directly to the columns in Table 3.23.

Age-agnostic classifier. The current classifier maps $X \to \hat{y}$, not $(X, \text{age}) \to \hat{y}$. BioBERT has no age feature, so Pediatrics is never a default destination for adult ASTS colours and paediatric TEWS tables remain out of scope. Before the pathway card is finalised, a deterministic (non-neural) check runs on the complaint X. Define the marker set

$$\mathcal{M}_{\text{child}} = \{\texttt{child}, \texttt{son}, \texttt{daughter}, \texttt{baby}, \texttt{infant}, \texttt{toddler}\}. \tag{3.34}$$

Normalise X to lowercase. If any element of $\mathcal{M}_{\text{child}}$ appears as a substring of X, override the destination field of $P(C_{\text{NLP}})$ to Pediatrics; otherwise keep the adult destination for that colour. This rule is applied after $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$ and does not change the colour itself. It is a **prototype routing heuristic only**, not a reliable way to identify paediatric patients: phrases such as "8-year-old with fever" or "girl with severe abdominal pain" need not match $\mathcal{M}_{\text{child}}$ and would keep the adult destination. Structured age (or age-aware destination) remains future work.

The pathway configuration below represents the authors' documented interpretation of observed workflow at KNUST University Hospital and published SATS rules; it should not be interpreted as an official clinical protocol unless formally approved by the hospital.

Colour	$T_{\max}$	Destination	Escalation
Red	0 min	Resuscitation / Surgery	Immediate senior alert
Orange	10 min	Internal Medicine / Surgery / Obs. & Gynae	Head nurse if unclaimed
Yellow	60 min	Outpatient urgent stream	Re-assess if timer expires
Green	< 4 h	Outpatient / Dental / Eye / Public & Preventive Health	Return only if worse

Table 3.23: SATS pathway protocol at KNUST University Hospital (clinical destinations; adult ASTS default).

Table 3.24 is the authoritative full map: one row per NLP outcome (including gate rejection), with primary clinical departments, support actions, escalation rules, and an example complaint phrase.

Scenario	Trigger	$T_{\max}$	Primary clinical departments	Support actions	Escalation
Gate reject	$g(X) < 0.30$	n/a	Hold at intake	Nursing: request complaint; HIT: no colour stored	Re-submit with clinical text
Red	$\hat{c} = 1$	0 min	Surgery, Internal Medicine	Nursing (vitals, TEWS), Pharmacy (emergency), Health Info (Red flag), Admin (bed bypass), HIT (Code Red)	Immediate senior clinician
Orange	$\hat{c} = 2$	10 min	Internal Medicine, Surgery, Obs. & Gynae	Nursing (monitor, timer), Diagnostics (stat labs), Pharmacy, Health Info, Admin, HIT	Head nurse if unclaimed
Yellow	$\hat{c} = 3$	60 min	Outpatient urgent, Diagnostics	Nursing (vitals, TEWS, timer), Pharmacy, Health Info, HIT, Accounts (defer billing)	Re-assess at 60 min
Green	$\hat{c} \in \{4, 5\}$	<4 h	Outpatient, Dental, Eye, Public & Preventive Health	Nursing (brief triage), Diagnostics (routine), Health Info, HIT, Accounts	Return if worse

Table 3.24: Complete NLP pathway scenarios at KNUST University Hospital (gate reject plus four SATS colours). Pediatrics is not a default Yellow destination; destination is overridden to Pediatrics only by the prototype lexical child-marker rule on $\mathcal{M}_{\text{child}}$ in X, which is not a reliable paediatric identifier.

Scenario	Example complaint phrase
Gate reject	"Hello, how are you?" (no clinical content)
Red	"Crushing chest pain, cannot breathe, sweating heavily"
Orange	"Severe abdominal pain in pregnancy, bleeding"
Yellow	*"I have a headache and feel feverish. My body aches and I feel weak."*
Green	"Routine dental pain for two weeks, no fever"

Table 3.25: Example chief complaints triggering each pathway scenario.

Table 3.26 maps all hospital departments to their role in the pathway system. Clinical departments receive patients by colour; support departments execute actions attached to each pathway card.

Department	**Role**	**Colours / function**
Internal Medicine	Clinical	Red, Orange (acute medical)
Surgery	Clinical	Red, Orange (trauma, acute surgical)
Obstetrics and Gynaecology	Clinical	Orange (obstetric urgency)
Pediatrics	Clinical	Child markers only (not default Yellow)
Outpatient	Clinical	Yellow (urgent), Green (routine)
Diagnostics Services	Clinical	Yellow, Green (parallel labs/imaging)
Dental	Clinical	Green
Eye	Clinical	Green
Public Health	Clinical	Green (community referrals)
Preventive Health	Clinical	Green (screening, wellness)
Nursing	Support	All colours (triage, vitals, reassessment)
Pharmacy	Support	All colours (medication on protocol)
Health Information	Support	All colours (record colour and timer)
Hospital Information Technology	Support	All colours (digital routing, API)
Administration	Support	All colours (bed and queue management)
Accounts	Support	Green, Yellow (billing after stabilisation)

Table 3.26: Department routing map at KNUST University Hospital.

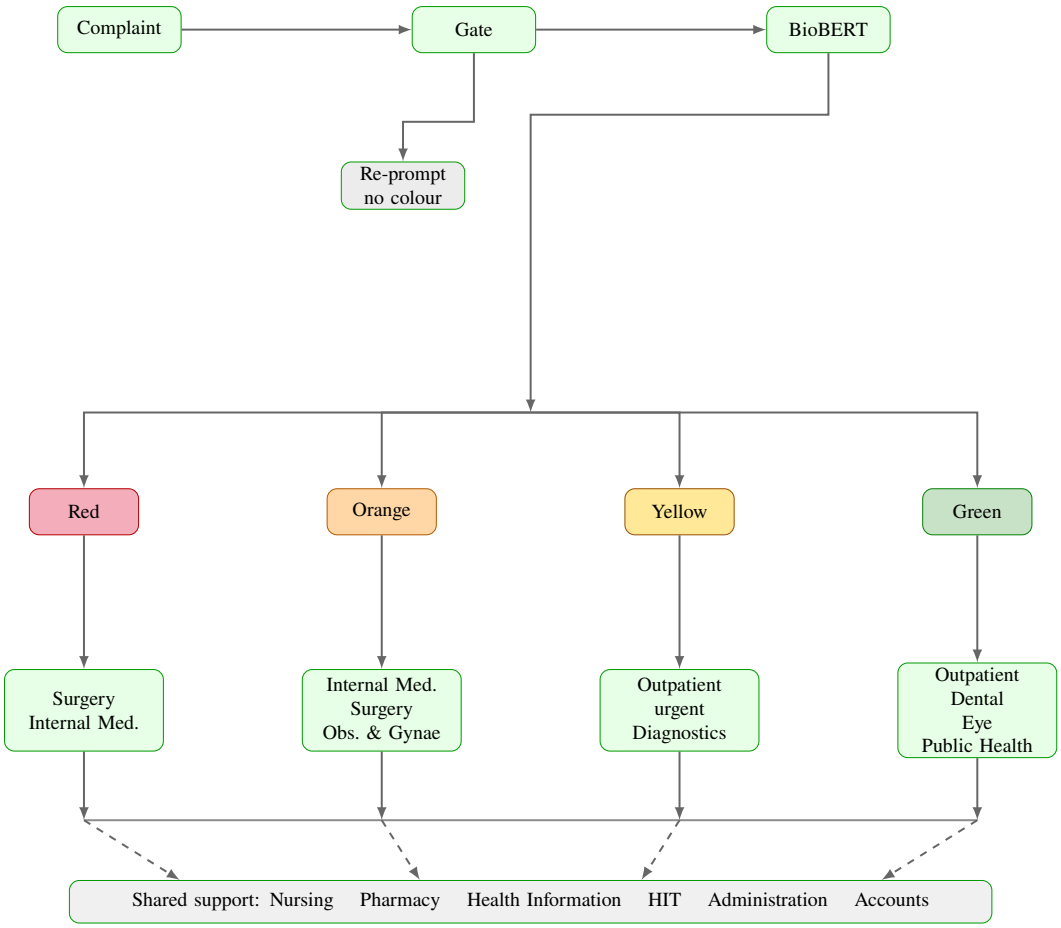


Figure 3.10: Complete NLP pathway map at KNUST University Hospital: gate rejection and all four SATS colours with clinical destinations and shared support services.

Remark 3.11. *Routing is protocol-driven; nurses may override the chatbot suggestion after clinical assessment. The chatbot supplies an initial colour and department direction before vitals are measured, which may reduce idle time at intake.*

Gate rejection. When $g(X) < 0.30$, the pipeline stops before BioBERT (Table 3.24). Nursing staff request a proper chief complaint; no colour or pathway card is issued and Health Information stores no triage record until valid clinical text is submitted.

Red ($\hat{c} = 1$). Immediate life-threatening presentations (e.g. crushing chest pain with dyspnoea) map to Red with $T_{\max} = 0$. Surgery and Internal Medicine activate resuscitation pathways; Nursing records immediate vitals and TEWS; Administration bypasses routine registration; Hospital Information Technology raises a Code Red alert.

Orange ($\hat{c} = 2$). Very urgent cases such as severe abdominal pain in pregnancy route to Internal Medicine, Surgery, or Obstetrics and Gynaecology within ten minutes. Nursing starts a countdown timer; Diagnostics may run stat laboratory work; escalation to the head nurse occurs if no clinician claims the patient within $T_{\max}$.

Yellow ($\hat{c} = 3$). The running illustrative complaint (*"I have a headache and feel feverish. My body aches and I feel weak."*) maps here at approximately 72% confidence. Outpatient urgent stream and parallel Diagnostics orders apply under the adult ASTS default; Pediatrics applies only when the lexical child-marker rule fires on X. Nursing sets a 60-minute reassessment timer and records the colour in Health Information.

Green ($\hat{c} \in \{4, 5\}$). Non-urgent presentations (e.g. routine dental pain without fever) stream to Outpatient, Dental, Eye, or Public and Preventive Health within four hours. The proposed routing is intended to divert low-acuity patients away from acute waiting areas; Accounts handles billing after brief Nursing triage.

3.15 Medical relevance gate

Running BioBERT on non-clinical text would waste compute and produce meaningless acuity scores. A lightweight rule-based gate scores clinical relevance before the deep model runs:

$$g(X) = \text{clip}\left(\sum_j s_j(X) - p(X),\ 0,\ 1\right), \tag{3.35}$$

where

- X is the English chief-complaint string;
- $\sum_j s_j(X)$ is an additive clinical evidence score assembled as follows (text-only path used in this project):
  - symptom lexicon: +0.15 per matched term, capped at 0.45;
  - triage or hospital-context phrases (e.g. OPD, nurse, chief complaint): +0.25 if present;
  - optional disease or drug entity bonuses when an external entity tagger is available;
- $p(X)$ is the non-clinical penalty: $p(X) = 0.5$ if a greeting-only, social-chatter, or other non-clinical pattern matches, and $p(X) = 0$ otherwise;
- $\text{clip}(u, 0, 1) = \max(0, \min(1, u))$ keeps the gate score in the unit interval $[0, 1]$.

Greeting and non-clinical patterns. The gate does not treat "greeting" as a vague category. Standalone greeting-only strings matched by the prototype include `hi`, `hello`, `hey`, `good morning`, `good afternoon`, and `good evening` (optional `there`, optional trailing punctuation). Social chatter combines a greeting opener with `how are you` or `what's up`. Separate non-clinical topic patterns cover sports (e.g. playing football),

shopping, school work, weather, travel, work chat, cooking, and entertainment. Any such match sets $p(X) = 0.5$.
In order: add the clinical evidence, subtract the non-clinical penalty, then clip into $[0, 1]$. Classification proceeds only if

$$g(X) \geq \tau, \qquad \tau = 0.30, \tag{3.36}$$

where τ is the acceptance threshold (overridable by environment configuration in the prototype).

Why $\tau = 0.30$. With the lexicon weights above, one symptom term yields 0.15, two terms yield 0.30, and three terms yield 0.45 (before the 0.45 cap binds). Setting $\tau = 0.30$ therefore admits short but clear complaints such as "chest pain" (two lexicon hits) while still rejecting greeting-only and sports chat, which score $g(X) = 0$ after the penalty. The older candidate $\tau = 0.35$ sits above two lexicon hits, so it systematically rejects two-term clinical phrases; that raises false clinical rejection risk and was therefore not retained as the project default.

Empirical gate validation. The threshold choice is supported by a hand-labeled contrast set of $N = 52$ English strings (28 clinical, 24 non-clinical), including multi-symptom complaints, short clinical phrases, lexicon-poor clinical wording (e.g. "I feel something strange in my chest"), named greetings, and social or sports chat. Treating clinical text as the positive class, define

$$\text{Sens} = \frac{TP}{TP + FN}, \qquad \text{Spec} = \frac{TN}{TN + FP},$$

together with accuracy, precision, and the confusion counts TP, FP, TN, FN. Here FN is a *false clinical rejection* (clinical text blocked before BioBERT) and FP is a *false clinical acceptance* (non-clinical text allowed through). Table 3.27 summarises both thresholds on this set.

τ	Acc	Prec	Sens	Spec	TP	FP	TN	FN
0.30 (deployed)	88.46%	100%	78.57%	100%	22	0	24	6
0.35 (candidate)	61.54%	100%	28.57%	100%	8	0	24	20

Table 3.27: Clinical relevance gate metrics on the $N = 52$ hand-labeled contrast set. Positive class = clinical. FN = false clinical rejection; FP = false clinical acceptance.

At the deployed $\tau = 0.30$, specificity and precision are 100% on this set (no false clinical acceptances among the 24 non-clinical strings), while sensitivity is 78.57% (6 false clinical rejections). Raising τ to 0.35 leaves specificity unchanged but collapses

sensitivity to 28.57% (20 false rejections), which is why 0.30 is preferred. Residual false rejections at 0.30 are mainly lexicon-poor clinical wording (e.g. "I feel something strange in my chest", "I am not feeling well") that never accumulate enough s_j. In those cases the gate fails closed for colour and asks for a richer chief complaint; nurse override and re-submission remain the safety net. This contrast set is illustrative, not a prospective audit of live KNUST University Hospital traffic: $N = 52$ is enough for a prototype threshold check, but not enough for broad claims about live gate performance.

Example input X	$g(X)$	Gate	Outcome
"chest pain"	0.30	Pass	Classify
"chest pain and fever since this morning"	≥ 0.30	Pass	Classify
"headache and feverish"	0.45	Pass	Classify
"I feel something strange in my chest"	0.15	Fail	Request clinical text
"Hello, how are you?"	0	Fail	Request clinical text
"good morning"	0	Fail	Request clinical text
"Going to play football"	0	Fail	Request clinical text

Table 3.28: Illustrative clinical relevance gate outcomes at the deployed threshold $\tau = 0.30$.

Example 3.7. *For "headache and feverish", the lexicon matches three overlapping terms (*`headache`*,* `fever`*,* `ache`*), giving*

$$\sum_j s_j = \min(0.45,\ 3 \times 0.15) = 0.45,$$
$$g(X) = \text{clip}(0.45 - 0,\ 0,\ 1) = 0.45 \geq 0.30,$$

so classification proceeds. For "Hello, how are you?" with no symptom hits and greeting or social-chatter penalty $p(X) = 0.5$:

$$\sum_j s_j = 0,$$
$$g(X) = \text{clip}(0 - 0.5,\ 0,\ 1) = 0 < 0.30,$$

and the system requests a chief complaint instead of running BioBERT.

3.16 Worked numerical walkthrough

This section traces one complaint through every stage of the NLP pipeline in Figure 3.1 and Section 3.17 with concrete numbers, so the Softmax and pathway sections are not

left abstract. Let

$$X = \text{“I have a headache and feel feverish. My body aches and I feel weak.”}$$

as elsewhere in the project.

Step 1: Gate. Symptom lexicon hits for the running complaint include `headache`, `fever/ache`, and `weak`, giving $\sum_j s_j = \min(0.45,\ 4 \times 0.15) = 0.45$ and $g(X) = 0.45 \geq 0.30$ with no greeting penalty (Section 3.15). Classification proceeds.

Step 2: Tokenisation. WordPiece yields a short sequence with `[CLS]`, symptom subwords, `[SEP]`, and pads to length $M = 128$. Only the first few positions carry clinical content; pads are masked in attention. The map τ is deterministic: the same string always produces the same IDs.

Step 3: Embeddings and $H^{(0)}$. Each of the M positions receives a 768-dimensional vector

$$H^{(0)}_{i,:} = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(0).$$

Shape $H^{(0)} \in \mathbb{R}^{128 \times 768}$: rows are tokens, columns are features (Section 3.8).

Step 4: Twelve encoder layers. Self-attention mixes information across rows. After layer $\ell = 12$, row 0 is the summary $h_{[\text{CLS}]} \in \mathbb{R}^{768}$. Schematic attention (Table 3.15) illustrates higher weight on symptom tokens than on fillers.

Step 5: Softmax. Using the illustrative logits of Section 3.11,

$$\hat{\mathbf{y}} \approx [0.03,\ 0.07,\ 0.69,\ 0.15,\ 0.05], \qquad \hat{c} = 3.$$

This is a Categorical distribution with mode Level 3.

Step 6: Colour and pathway.

$$C_{\text{NLP}} = f_{\text{SATS}}(3) = \text{Yellow}.$$

The pathway card is

$$P(\text{Yellow}) = \big(60\,\text{min},\ \text{Outpatient urgent stream},\ \text{actions},\ \text{escalation}\big)$$

as in Table 3.24. Nursing sets a 60-minute reassessment timer; Diagnostics may run in parallel.

Step 7: Parallel TEWS (when vitals arrive). Suppose later measurements give TEWS integer $T = 4$. Then C_{TEWS} is Yellow on the SATS vital bands (Section 3.3). Text colour and vital colour agree in this illustration; if they disagreed, nurse judgment and SATS discriminators govern, not automatic Softmax override.

The walkthrough shows why every mathematical object in Chapter 3 appears in the deployed API response: gate metadata, $\hat{\mathbf{y}}$, C_{NLP}, and pathway fields.

3.17 End-to-end summary

For any chief complaint X, the implemented text path (matching the prototype `/predict` endpoint) runs only when the clinical gate passes: $g(X) \geq \tau_g$ with deployed $\tau_g = 0.30$ (Section 3.15); otherwise the system stops after the gate and assigns no colour. When the gate passes,

$$\begin{aligned} X \xrightarrow[3.5]{\tau} \text{ids} \xrightarrow[3.8]{\text{emb}} H^{(0)} \xrightarrow[3.9--3.10]{B} h_{[\text{CLS}]} \\ \xrightarrow[3.11]{S} \hat{\mathbf{y}} \xrightarrow{\arg\max} \hat{c} \xrightarrow[3.13]{f_{\text{SATS}}} C_{\text{NLP}} \xrightarrow[3.14]{P} P(C_{\text{NLP}}), \end{aligned} \tag{3.37}$$

where τ is WordPiece tokenisation to integer IDs, emb builds the input embedding matrix $H^{(0)}$, B is the twelve-layer BioBERT map to the summary vector $h_{[\text{CLS}]}$, S is the Softmax classification head producing the five-class distribution $\hat{\mathbf{y}}$, $\hat{c} = \arg\max_c \hat{y}_c$ is the predicted ESI-style level, $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$ is the SATS-aligned colour, and $P(C_{\text{NLP}})$ is the pathway card. The tokeniser map τ is distinct from the gate threshold τ_g and the confidence threshold τ_c below.

Composite operator form. Writing the maps explicitly for a mathematics audience, let $M = 128$ be the project sequence length, V the WordPiece vocabulary size, and

$$\begin{aligned} g &: X \mapsto [0,1] \quad \text{(clinical gate score)}, \\ \tau &: X \mapsto \{1,\dots,V\}^M \quad \text{(padded WordPiece IDs)}, \\ \text{emb} &: \{1,\dots,V\}^M \mapsto \mathbb{R}^{M\times 768} \quad \text{(builds } H^{(0)}\text{)}, \\ B &: \mathbb{R}^{M\times 768} \mapsto \mathbb{R}^{768} \quad (L = 12 \text{ encoder layers to } h_{[\text{CLS}]}), \\ S &: \mathbb{R}^{768} \mapsto \Delta^4 \quad \text{(Softmax head)}, \\ \arg\max &: \Delta^4 \mapsto \{1,\dots,5\} \quad \text{(predicted level } \hat{c}\text{)}, \\ f_{\text{SATS}} &: \{1,\dots,5\} \mapsto \{R,O,Y,G\}, \\ P &: \{R,O,Y,G\} \mapsto \text{PathwayCards}, \end{aligned}$$

where the probability simplex is

$$\Delta^4 = \{p \in \mathbb{R}^5 : p_i \geq 0,\ \sum_{i=1}^{5} p_i = 1\}.$$

In informal prose the embedding step may be absorbed into the "encoder stack," but the typed composition keeps emb explicit so $H^{(0)}$ is not orphaned. The text-path operator (when the gate passes) is

$$F = P \circ f_{\text{SATS}} \circ \arg\max \circ S \circ B \circ \text{emb} \circ \tau, \tag{3.38}$$

subject to $g(X) \geq \tau_g$. Empirically scored evaluation applies to $S \circ B \circ \text{emb} \circ \tau$ (ESI-style $\hat{\mathbf{y}}$); f_{SATS} and P are proposed or protocol maps in the hierarchy of evidence (Chapter 1).

Confidence deferral. Let $\hat{y}_{(1)} = \max_c \hat{y}_c$ be the largest Softmax probability and $\hat{y}_{(2)}$ the second-largest. The operational nurse-defer rules are

$$\max_c \hat{y}_c < \tau_c \quad \Rightarrow \quad \text{mandatory nurse review,} \tag{3.39}$$

$$\hat{y}_{(1)} - \hat{y}_{(2)} < \delta \quad \Rightarrow \quad \text{mandatory nurse review,} \tag{3.40}$$

with prototype policy parameters $\tau_c = 0.60$ and $\delta = 0.10$ (not clinical-optimised cutoffs). Either condition triggers review before the pathway card is treated as actionable; Softmax scores remain uncalibrated model probabilities, not clinical certainties.

1. **Clinical relevance gate.** Compute $g(X)$; if $g(X) < \tau_g = 0.30$, return a request for clinical detail and stop.

2. **Tokenisation and embedding.** Apply τ then emb to build $H^{(0)} \in \mathbb{R}^{M \times 768}$ with $M = 128$ (Sections 3.5–3.8).

3. **BioBERT encoding.** Apply B with $L = 12$ encoder layers and hidden size 768 (Sections 3.9–3.10) to obtain $h_{[\text{CLS}]} \in \mathbb{R}^{768}$.

4. **Classification.** Apply the Softmax head S (Section 3.11) with learned parameters $W \in \mathbb{R}^{5 \times 768}$ and $\mathbf{b} \in \mathbb{R}^5$:

$$\hat{\mathbf{y}} = \text{softmax}(W h_{[\text{CLS}]} + \mathbf{b}),$$
$$\hat{c} = \arg\max_c \hat{y}_c.$$

5. **Colour and pathway.** Apply $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$ and return pathway card $P(C_{\text{NLP}})$ (Sections 3.13–3.14).

The illustrative complaint in this chapter instantiates the same pipeline; one worked outcome is level 3 (Yellow) at 72% confidence for headache and feverish symptoms. TEWS (Section 3.3) remains the nurse-facing vital-sign score when measurements become available.

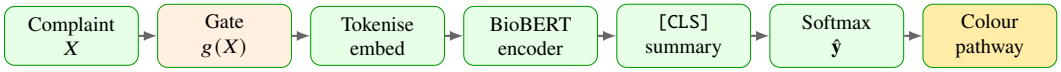


Figure 3.11: Compact end-to-end NLP triage path (TEWS documented separately in Section 3.3).

The diagram above matches the pipeline in Section 3.1 and Equation (3.37). In order:

1. **Complaint** X enters as free text from the patient or attendant.
2. **Gate** computes $g(X)$; if $g(X) < \tau_g$, non-clinical input is rejected before BioBERT runs.
3. **Tokenise and embed** apply τ and emb to map text to integer IDs and build $H^{(0)}$.
4. **BioBERT** applies B ($L = 12$ encoder layers with self-attention) across all tokens.
5. **[CLS] summary** extracts the fixed-length vector $h_{[\mathrm{CLS}]} \in \mathbb{R}^{768}$ from row 0.
6. **Softmax** applies S to output acuity probabilities $\hat{\mathbf{y}}$ over five classes; $\hat{c} = \arg\max_c \hat{y}_c$.
7. **Colour and pathway** apply $f_{\mathrm{SATS}}(\hat{c})$ to obtain C_{NLP} and return the pathway card $P(C_{\mathrm{NLP}})$.

3.18 Implementation note

Figure 3.12 shows the deployable system: patient-facing web interface, chatbot API, and hospital routing layer. The NLP mathematics in Figures 3.1 and 3.11 execute inside the API; department routing applies facility pathway rules after classification.

The prototype exposes a chatbot application programming interface (API) endpoint: JSON **English** chief complaint in; acuity probabilities, C_{NLP}, pathway fields, and gate metadata out. Model weights and tokenizer files ship with the repository listed in Appendix A. This project treats the mathematics above as the specification; measured results appear in Chapter 4.

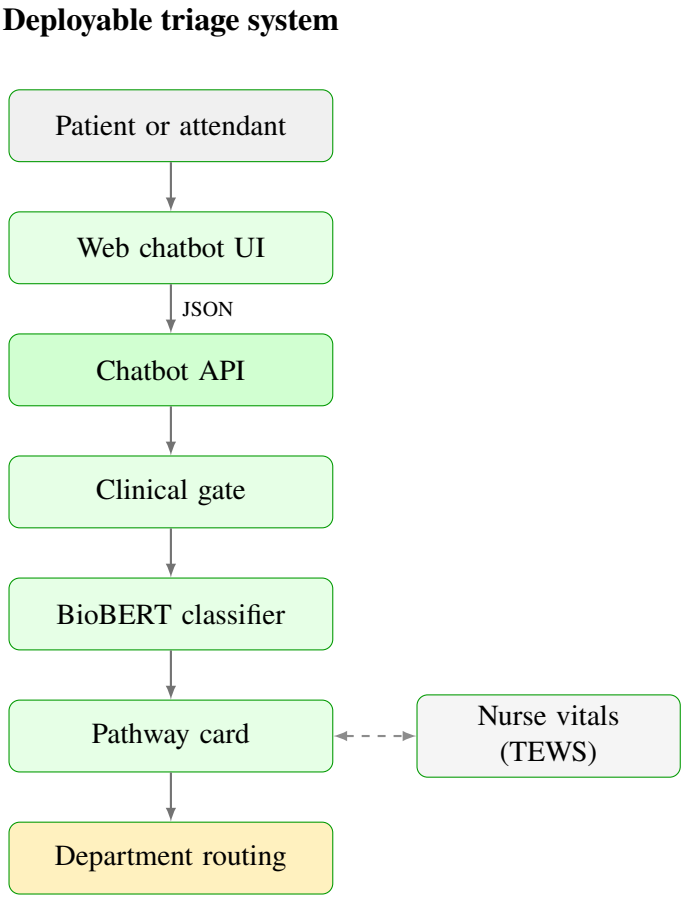


Figure 3.12: System architecture: user interface, chatbot API, BioBERT inference pipeline, pathway card, and department routing. Nurse TEWS runs in parallel (dashed); prototype code in Appendix A.

CHAPTER 4

RESULTS

Evaluating the BioBERT triage classifier requires a structured analysis of predictive performance under domain shift and pathway implications. Architecture, TEWS, and the three training stages are specified in Chapter 3 (Sections 3.9, 3.3, 3.12). Primary reported metrics come from **external** evaluation on the National Hospital Ambulatory Medical Care Survey (NHAMCS) 2018–22 and the Medical Information Mart for Intensive Care IV Emergency Department (MIMIC-IV-ED) open demo matching the v2.2 triage schema (National Center for Health Statistics (NCHS), 2024; Johnson et al., 2023b,a). Triagegeist is used for training and validation only. None of these sources is prospective live performance at KNUST University Hospital, and the model has not yet been clinically tested on real patients there.

4.1 Deployed checkpoints and evaluation setup

The deployed classifier is the Stage 3 fine-tuned model. Encoder architecture, TEWS, Stages 1–3, and Hugging Face identifiers are in Sections 3.9, 3.3, and 3.12 (Table 3.18). External metrics follow in Section 4.2.

Figure 4.1 shows the project inference configuration ($M = 128$); see Section 3.9 and Lee et al. (2020) for BioBERT-base constants.

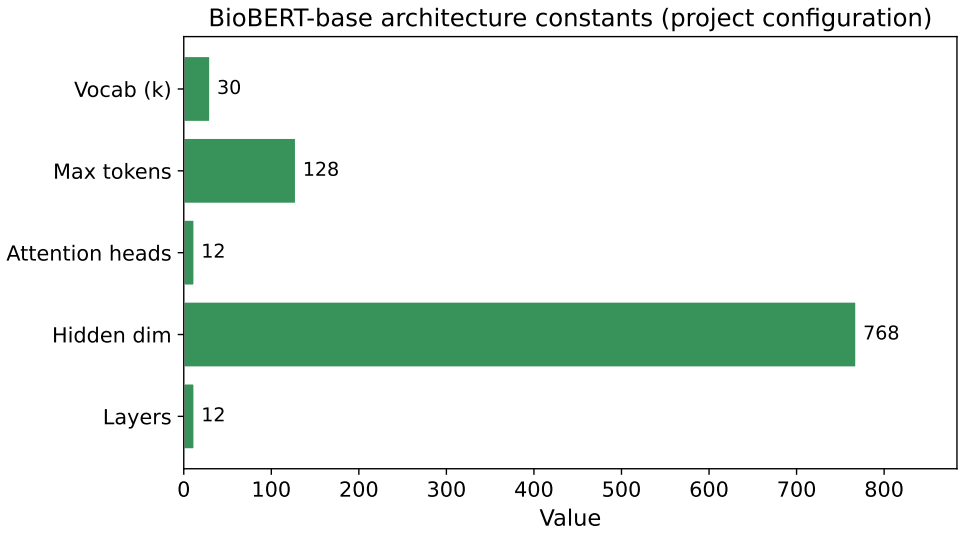


Figure 4.1: Project inference configuration constants ($M = 128$); architecture in Section 3.9.

4.2 Fine-tuned model analysis

This chapter reports **external** predictive metrics, colour accuracy via f_{SATS}, under/over-triage, and pathway implications.

4.2.1 Evaluation protocol

Primary reported metrics are from external evaluation on MIMIC-IV-ED and NHAMCS. Triagegeist is used for training/validation only. The split was designed to reduce the risk of data leakage between Stage 3 training and published scores. None of these sources is prospective live performance at KNUST University Hospital, and the model has not yet been clinically tested on real patients there.

1. **Train / validation (Triagegeist).** Stratified 90/10 split (seed 42) for Stage 3 fine-tuning and checkpoint selection. No Triagegeist holdout accuracy is reported as a Results headline.

2. **A: MIMIC-IV-ED.** Text-only on `chiefcomplaint` $\rightarrow$ predict acuity; gold = `acuity` $\in \{1, \ldots, 5\}$. Full MIMIC-IV-ED v2.2 is credentialed on PhysioNet (Johnson et al., 2023a); this project evaluates the open demo matching the same triage schema (Johnson et al., 2023b) (N = 207 after filters) until a full extract is available.

3. **B: NHAMCS 2018–22.** Text-only on `chief_complaint_text` $\rightarrow$ predict acuity; gold = `target_triage_acuity` (National Center for Health Statistics (NCHS), 2024). After filters, N = 58,124 rows remain; metrics below use a stratified subsample of N = 12,000 (seed 42) for compute, with full-set class support reported in Table 4.1.

Pre-processing (both external sets). Drop empty, NaN, or whitespace-only complaints; keep acuity in $\{1, \ldots, 5\}$; do *not* expand nursing abbreviations (e.g. CP, SOB); tokenise with BioBERT WordPiece and truncate/pad to project length M = 128.
Two checkpoints are compared:

1. **Baseline fine-tuned BioBERT** (project default): end-to-end fine-tuning without minority rebalancing beyond the natural train distribution.

2. **Stratified-oversampling variant:** text augmentation plus random minority oversampling on the Triagegeist training split, then end-to-end fine-tuning (Section 4.2.5).

Metrics: five-class accuracy and macro-F1; 4-colour accuracy after f_{SATS}; under-triage (predicted less urgent than true); over-triage (more urgent); critical under-triage (true Red→non-Red; true Orange→Yellow/Green); ordinal mean absolute error (MAE); latency.

Accuracy terminology. These quantities must not be blurred. **Five-class (text-classification) accuracy** is exact ESI-style level match. **Colour accuracy** is match after f_{SATS} (Levels 4 and 5 both map to Green). **Gate accuracy** is clinical versus non-clinical discrimination on the contrast set (Section 3.15), not acuity classification. **Clinical or pathway correctness** is not claimed from offline five-class or colour accuracy alone.

Error taxonomy. Classification, colour, and under-triage errors are related but not identical. For gold level y and prediction $\hat{y}$,

$$\begin{aligned} E_{\text{class}} &= \mathbf{1}(\hat{y} \neq y), \\ E_{\text{colour}} &= \mathbf{1}\big(f_{\text{SATS}}(\hat{y}) \neq f_{\text{SATS}}(y)\big), \\ E_{\text{under}} &= \mathbf{1}(\hat{y} > y) \end{aligned}$$

(with ESI convention that smaller integers are more urgent). Ordinal severity of class error is summarised by

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i|.$$

Cost-sensitive objective (interpretive). Stakeholders may weigh under-triage, over-triage, and classification error differently. An interpretive risk functional (not used as the training loss in this project) is

$$R(\theta) = \lambda_u U(\theta) + \lambda_o O(\theta) + \lambda_e E(\theta),$$

where U is under-triage rate, O is over-triage rate, E is five-class error rate, and $\lambda_u, \lambda_o, \lambda_e \geq 0$ encode relative clinical cost. Stratified oversampling is interpreted as increasing emphasis on rare Red recall at the expense of E and often O.

4.2.2 Headline external metrics

Table 4.2 is the primary results table for this chapter (NHAMCS and MIMIC demo; baseline vs stratified oversampling). Supporting acuity counts appear in Table 4.1.

4.2.3 Acuity distribution on test sets

Table 4.1 reports gold acuity counts by level after the shared pre-processing in Section 4.2.1. The NHAMCS counts come from the 2018–22 US emergency-department survey text field `chief_complaint_text` with gold labels `target_triage_acuity` (National Center for Health Statistics (NCHS), 2024): $N = 58{,}124$ rows after filters, with the stratified evaluation subsample ($N = 12{,}000$) used for published metrics. The MIMIC counts come from the open MIMIC-IV-ED demo on PhysioNet (Boston ED phrasing), mapping `chiefcomplaint` to gold `acuity` (Johnson et al., 2023b) ($N = 207$ after filters). Real emergency departments are imbalanced: NHAMCS is dominated by Levels 3–4, and Level 1 is scarce (846 of 58,124; 1.5%). MIMIC demo counts are small, especially Levels 4–5, so rare-class metrics there are interpreted cautiously.

Set	N_1	N_2	N_3	N_4	N_5	**Total**
NHAMCS (full retained)	846	8,597	29,568	16,715	2,398	58,124
NHAMCS (eval subsample)	175	1,775	6,104	3,451	495	12,000
MIMIC-IV-ED demo	18	97	90	2	0	207

Table 4.1: Gold acuity counts by level on the NHAMCS and MIMIC-IV-ED demo test sets after pre-processing.

4.2.4 Why prior in-distribution accuracy looked inflated

Let $N = 80{,}000$ denote the Triagegeist training-pair count used for Stage 3. After normalising complaint strings, only

$$N_{\text{unique}} = 4{,}949$$

distinct complaints remain, so the duplicate rate is

$$D = 1 - \frac{N_{\text{unique}}}{N} = 1 - \frac{4{,}949}{80{,}000} \approx 0.9381$$

(about 93.8%; 4,184 templates appear at least five times). Random row-level splitting can therefore place near-identical linguistic templates in both training and validation folds, inflating in-distribution accuracy. High template reuse helps explain why earlier holdout figures near 99.9% looked unrealistically strong: they largely measure agreement on repeated synthetic phrasing, not generalisation. Those figures are therefore retired from the Results headline in favour of MIMIC/NHAMCS metrics below. A full complaint-level deduplicated retrain is left as future work.

4.2.5 Stratified oversampling for class imbalance

Stage 3 training on Triagegeist is class-imbalanced: Level 3 (Urgent) is the mode, while Level 1 (Resuscitation / Red) is the minority. Rebalancing the training split can raise Red recall, but often at a cost to precision and overall accuracy. Two natural ways to reduce that imbalance were available for this project.

A: Classical embedding SMOTE. Raw chief-complaint strings are discrete, so SMOTE cannot interpolate between token sequences directly (Chawla et al., 2002). One workable route is to freeze (or otherwise detach) BioBERT, extract fixed feature vectors such as $h_{[\mathrm{CLS}]} \in \mathbb{R}^{768}$, apply SMOTE in that continuous space to synthesise minority examples, and train a *separate* linear classification head on the resulting embedding set. Gradients never update the text encoder during that second stage.

B: Stratified oversampling (chosen). Keep the end-to-end Stage 3 pipeline of Chapter 3: WordPiece tokenisation, BioBERT encoder, and Softmax head trained jointly. On the Triagegeist *training* split only, apply text augmentation to minority urgent classes, then random minority oversampling toward configured target counts, and continue full encoder-plus-head updates. The training configuration sets `USE_EMBEDDING_SMOTE=False`, so embedding-space SMOTE is not part of the published second checkpoint.

This project adopts **B** for the published imbalance-mitigated checkpoint and does *not* report a classical embedding-SMOTE model as the official second result.

The choice follows from the training objective. End-to-end Stage 3 is meant to adapt WordPiece and attention features to short chief-complaint language; that requires backpropagation into BioBERT. SMOTE-generated points in embedding space cannot be pushed back through the text encoder unless BioBERT is frozen, which would abandon that adaptation goal and replace it with a shallow head on static features. Stratified oversampling instead keeps real or lightly paraphrased complaint text inside the same tokenizer → encoder → Softmax path already specified in Chapter 3. Calling B "SMOTE" would therefore be inaccurate for the panel: Chawla-style SMOTE (Chawla et al., 2002) describes A, which was considered but not used for the published checkpoint. The NHAMCS and MIMIC comparisons below therefore contrast the baseline Stage 3 model with this stratified-oversampling checkpoint, not with an embedding-SMOTE head.

4.2.6 NHAMCS results

Table 4.2 summarises headline metrics on the stratified NHAMCS subsample (N = 12,000) and, for comparison in the same panel, the MIMIC-IV-ED demo (N = 207).

Metric	NHAMCS base.	NHAMCS overs.	MIMIC base.	MIMIC overs.
N	12,000	12,000	207	207
Accuracy	0.4760	0.3045	0.3043	0.2126
Macro-F1	0.2882	0.2146	0.1748	0.1781
Colour accuracy	0.4995	0.3422	0.3043	0.2126
MAE (ordinal)	0.6272	0.9335	1.0000	1.0773
Under-triage	0.2839	0.3903	0.6377	0.6087
Over-triage	0.2401	0.3052	0.0580	0.1787
Recall L1 (Red)	0.0800	0.2171	0.1111	0.5000
Precision L1	0.8235	0.0597	0.4000	0.2647
Latency p50 (ms)	94.00	96.18	n/a	n/a

Table 4.2: Classification, colour, ordinal MAE, under/over-triage, and latency metrics. MIMIC columns use the open demo ($N = 207$). MIMIC latency is reported as n/a: published timing uses the NHAMCS evaluation run (Section 4.2.9).

Reading Table 4.2 (point estimates). On NHAMCS, baseline BioBERT reaches 47.60% five-class accuracy, 0.288 macro-F1, and 49.95% colour accuracy after f_{SATS}. Colour accuracy sits slightly above five-class accuracy because Levels 4 and 5 both map to Green. Ordinal MAE is 0.627 at baseline and rises to 0.934 after stratified oversampling: when the model errs under rebalancing, the average distance on the ESI ladder widens even while Red recall improves. Under-triage is 28.39% and over-triage 24.01% at baseline; both rise under oversampling (to 39.03% and 30.52%), so the sensitivity gain is not free on directional error. Level 1 recall is only 8.00% (support 175) despite 82.35% Level 1 precision: most true Red cases are predicted as Orange or Yellow rather than Red, which is the primary safety concern under domain shift (Berkowitz et al., 2019). Oversampling lifts Red recall to 21.71% but collapses Level 1 precision to 5.97% and five-class accuracy to 30.45%: more true Reds are caught at the cost of many false urgent alerts. Median latency remains about 94–96 ms on NHAMCS, well under one second. MIMIC demo columns show the same qualitative pattern at smaller N (lower Acc, higher under-triage at baseline, Red recall rising under oversampling); that demo is interpreted fully in Section 4.2.7. Sampling uncertainty for the key Acc / colour Acc / Red-recall estimates is reported immediately before the confusion matrices below (Table 4.4), because those intervals are built from the same matrix cell counts.

Simple baselines on the same NHAMCS subsample. A majority-class predictor (always Level 3) reaches 50.87% accuracy and 0.135 macro-F1 with Level 1 recall 0 on the $N = 12{,}000$ NHAMCS subsample. Baseline BioBERT (trained on Triagegeist, evaluated under domain shift) is slightly below majority accuracy (47.60%) but

substantially above majority macro-F1 (0.288) and recovers non-zero Red recall (8.00%). A same-domain TF-IDF + logistic regression baseline trained on 70% of the NHAMCS subsample and tested on the held-out 30% reaches 45.83% accuracy, 0.346 macro-F1, and 28.85% Level 1 recall: lexical features help when train and test share NHAMCS phrasing, which is a different regime from Triagegeist→NHAMCS transfer. These comparisons answer whether BioBERT adds value beyond naive rules: it does on rare-class and macro metrics under shift, while raw accuracy alone can favour majority guessing on a Level 3-dominated survey. Table 4.3 reports the full computed panel, including MIMIC demo transfer rows.

Protocol caveat. Not a matched head-to-head: TF-IDF+LR is trained and tested within NHAMCS; BioBERT is trained on Triagegeist and tested on NHAMCS. The rows below are therefore not a fair same-data bake-off.

Corpus	Model	Acc	Macro-F1	MAE	Under	Over	L1 rec.
NHAMCS	Majority (L3)	0.5087	0.1349	0.5472	0.1625	0.3288	0.0000
NHAMCS	TF-IDF+LR	0.4583	0.3465	0.7344	0.2539	0.2878	0.2885
NHAMCS	BioBERT base.	0.4760	0.2882	0.6272	0.2839	0.2401	0.0800
MIMIC demo	Majority (L2)	0.4686	0.1595	0.5411	0.0870	0.4444	0.0000
MIMIC demo	TF-IDF+LR	0.2609	0.1824	1.5169	0.6280	0.1111	0.1111
MIMIC demo	BioBERT base.	0.3043	0.1748	1.0000	0.6377	0.0580	0.1111

Table 4.3: **Not a matched head-to-head.** TF-IDF+LR is same-domain within NHAMCS; BioBERT is Triagegeist→NHAMCS transfer. NHAMCS majority and BioBERT use the full stratified subsample (N = 12,000); TF-IDF+LR is a 70%/30% holdout (N_{test} = 3,600). MIMIC demo (N = 207): majority is constant Level 2; TF-IDF+LR is trained on the NHAMCS subsample and evaluated on MIMIC; BioBERT is Triagegeist-trained under domain shift (Table 4.2). Protocols differ across rows, so comparisons are interpretive rather than a single matched bake-off.

Significance of the baseline and ordinal metrics. Table 4.3 makes the accuracy-versus-safety gap concrete. On NHAMCS, majority accuracy (0.5087) exceeds BioBERT baseline accuracy (0.4760) because Level 3 dominates the survey subsample, yet majority Level 1 recall is exactly 0: a constant Level 3 rule never flags Red language. Raw accuracy alone is therefore a poor success measure for triage assist.

Macro-F1 and Level 1 recall tell a different story. BioBERT under Triagegeist→NHAMCS shift raises macro-F1 from 0.1349 (majority) to 0.2882 and recovers non-zero Red recall (0.0800). The same-domain TF-IDF+LR holdout reaches Level 1 recall 0.2885 with macro-F1 0.3465, but that model trains and tests on NHAMCS phrasing; it is not evidence that bag-of-words beats BioBERT under the transfer regime this project actually deploys. The table columns should be read with those protocols in mind.

Ordinal MAE is the mean absolute class distance $|\hat{y} - y|$. Majority MAE on NHAMCS is 0.5472; BioBERT baseline MAE is 0.6272, so when BioBERT errs it can land farther from the true ESI level even while improving rare-class detection relative to constant Level 3. Stratified oversampling widens that distance further (MAE 0.9335 in Table 4.2) while lifting Red recall, which is the same trade-off summarised under Finding 3 and the interpretive risk $R(\theta)$.

The published under-triage, over-triage, and five-class error rates are the empirical averages of the indicators E_{under}, over-triage, and E_{class} defined earlier in this chapter; colour mismatches after f_{SATS} correspond to E_{colour}. No separate taxonomy experiment was run beyond those rates. Likewise $R(\theta) = \lambda_u U + \lambda_o O + \lambda_e E$ remains interpretive: oversampling is read as raising emphasis on Red recall at the expense of classification error E and often over-triage O, not as an optimised schedule for the λ weights.

On the MIMIC demo ($N = 207$), majority accuracy remains high (0.4686) with MAE 0.5411, while NHAMCS-trained TF-IDF+LR and Triagegeist-trained BioBERT both show much larger ordinal error (MAE 1.5169 and 1.0000) and high under-triage. That pattern reinforces how fragile lexical and neural transfer are on a small, shifted ED corpus: headline accuracy for a constant label can look competitive while safety-relevant distance and under-triage degrade.

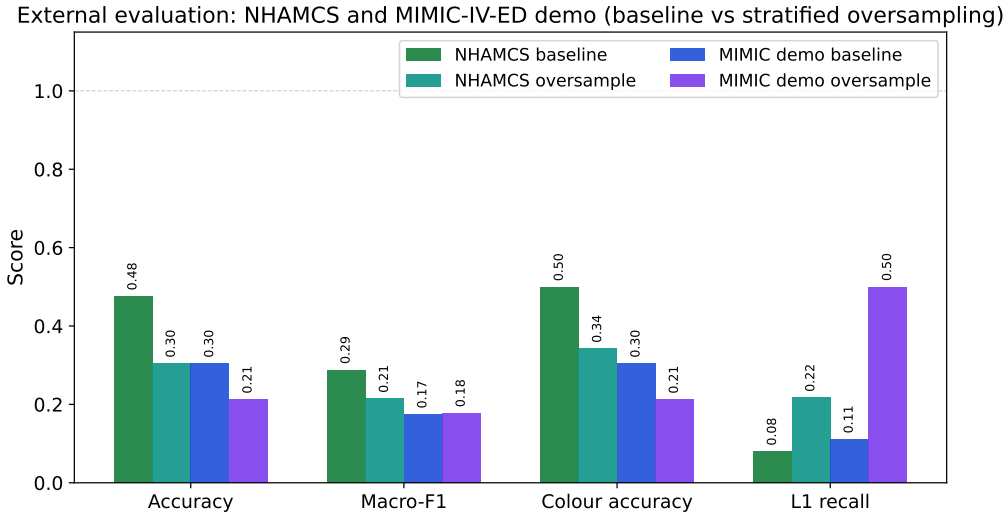


Figure 4.2: External headline metrics on NHAMCS (N = 12,000 stratified subsample) and MIMIC-IV-ED demo (N = 207): baseline versus stratified oversampling.

Figure 4.2 plots the same accuracy, macro-F1, colour accuracy, and Level 1 recall values as Table 4.2, so the baseline–oversample contrast is readable before the bootstrap intervals and confusion matrices below.

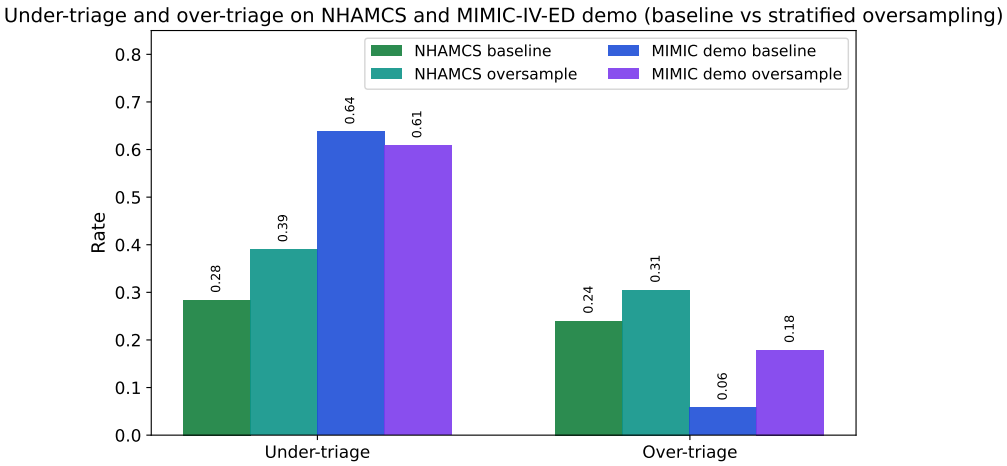


Figure 4.3: Under-triage and over-triage rates on NHAMCS (N = 12,000 stratified subsample) and MIMIC-IV-ED demo (N = 207): baseline versus stratified oversampling.

Under-triage (predicted less urgent than gold) is the safety failure mode: urgent language is routed too late. Over-triage (predicted more urgent than gold) is false escalation that loads acute streams and staff. Figure 4.3 shows that on NHAMCS, stratified oversampling raises *both* under-triage (28.39% to 39.03%) and over-triage (24.01% to 30.52%) even while Level 1 recall rises (Figure 4.2, Table 4.2), so sensitivity gains are not a free improvement in directional error. On the MIMIC demo, baseline under-triage is already high (63.77%); oversampling mainly increases over-triage (5.80% to 17.87%). These patterns reinforce mandatory nurse override under domain shift (Berkowitz et al., 2019).

Bootstrap uncertainty (what, why, and how to read the intervals). The intervals below are derived from the same confusion-matrix cell counts that the figures immediately after expand into readable error patterns. Point estimates in Table 4.2 do not show sampling uncertainty. A **bootstrap 95% confidence interval** answers a different question: if we resample this evaluation set many times, in what range does the metric usually fall? It is *not* a claim of clinical certainty at KNUST University Hospital.

What was computed. For each published confusion matrix, the cell counts were expanded into N pairs $(y, \hat{y})$. Then $B = 2000$ ordinary bootstrap resamples (with replacement, seed 42) recomputed five-class accuracy, colour accuracy after f_{SATS}, and Level 1 (Red) recall. The 2.5th and 97.5th percentiles of those B values form the percentile 95% CI in Table 4.4.

Why bootstrap, and why NHAMCS is primary. Level 1 is scarce (175 true Reds on the NHAMCS subsample; only 18 on the MIMIC demo), so a few additional hits or misses move Red recall substantially. Oversampling also claims a clear accuracy drop and a Red-recall rise; those claims should be checked against sampling noise rather than

two-decimal point estimates alone. The **detailed** reading centres on NHAMCS because it is the main published evaluation set (N = 12,000). The MIMIC open demo (N = 207) is a secondary transfer check: the same bootstrap procedure was applied, but with so few Reds the intervals are very wide, so MIMIC CIs document uncertainty on that small set rather than support precise sensitivity claims.

Checkpoint	Metric	Estimate	95% bootstrap CI
NHAMCS baseline	Five-class Acc	0.4760	[0.4670, 0.4847]
NHAMCS baseline	Colour Acc	0.4995	[0.4903, 0.5082]
NHAMCS baseline	Recall L1 (Red)	0.0800	[0.0435, 0.1236]
NHAMCS overs.	Five-class Acc	0.3045	[0.2960, 0.3132]
NHAMCS overs.	Colour Acc	0.3423	[0.3336, 0.3513]
NHAMCS overs.	Recall L1 (Red)	0.2171	[0.1579, 0.2784]
MIMIC demo base.	Five-class Acc	0.3043	[0.2464, 0.3671]
MIMIC demo base.	Colour Acc	0.3043	[0.2464, 0.3671]
MIMIC demo base.	Recall L1 (Red)	0.1111	[0.0000, 0.2667]
MIMIC demo overs.	Five-class Acc	0.2126	[0.1594, 0.2705]
MIMIC demo overs.	Colour Acc	0.2126	[0.1594, 0.2705]
MIMIC demo overs.	Recall L1 (Red)	0.5000	[0.2631, 0.7334]

Table 4.4: Bootstrap 95% CIs (B = 2000, seed 42) for NHAMCS (N = 12,000) and the MIMIC-IV-ED demo (N = 207). Primary interpretation is NHAMCS; MIMIC rows document uncertainty on the small secondary transfer set. Intervals quantify sampling uncertainty on each evaluation set, not clinical certainty at KNUST University Hospital.

Significance on NHAMCS. Five-class accuracy for the baseline is 0.476 with a tight interval [0.467, 0.485], reflecting the large sample. The oversampled checkpoint falls to 0.305 [0.296, 0.313]; the two intervals do not overlap under this resampling scheme, so the accuracy drop is stable, not a fluke of one draw. Colour accuracy follows the same pattern (baseline about 0.500 versus oversample about 0.342), and colour accuracy remains slightly above five-class accuracy because Levels 4 and 5 both map to Green under f_{SATS}. Red recall is the clinically important but noisy quantity: baseline 0.080 sits in a wide interval [0.043, 0.124] because only 175 true Level 1 cases are available. Oversampling raises the point estimate to 0.217 [0.158, 0.278]. Those Red-recall intervals do not overlap the baseline interval here, which supports a real sensitivity gain for RQ4 / Finding 3, yet the absolute level remains modest (even the upper end stays below about 0.28). The bootstrap therefore strengthens an honest trade-off story: oversampling moves Red recall up and overall accuracy down in a resampling-stable way, while still leaving critical-case detection too weak for unsupervised Red screening.
Significance on the MIMIC demo. On the open demo, baseline five-class (and colour) accuracy is 0.304 [0.246, 0.367]; oversampling lowers it to 0.213 [0.159, 0.271]. Red recall moves from 0.111 [0.000, 0.267] to 0.500 [0.263, 0.733]. Because only 18 true

Level 1 cases exist, the Red intervals span from near zero to high values: a few flips change the estimate dramatically. MIMIC therefore confirms that the evaluation pipeline runs on real Boston ED phrasing and that oversampling can inflate Red hits on a tiny support, but it does *not* justify precise claims about critical-case sensitivity. That is why the narrative weight stays on NHAMCS, with MIMIC reported for completeness. The confusion matrices next show *where* those errors sit on the five-class grid.

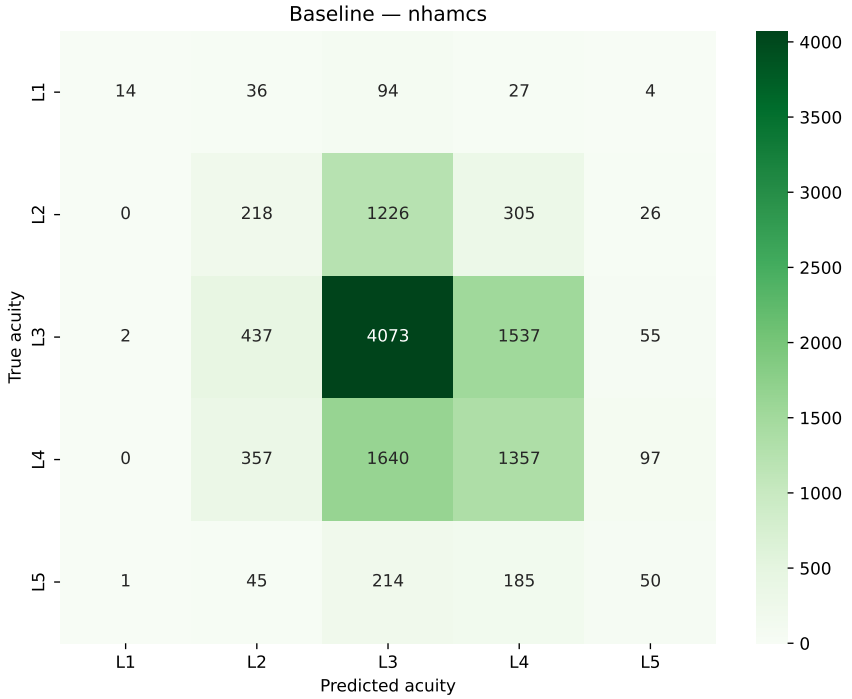


Figure 4.4: Baseline BioBERT confusion matrix on NHAMCS stratified subsample (N = 12,000). Rows = true acuity; columns = prediction. Dark diagonal cells are correct predictions; off-diagonal mass is triage error.

The NHAMCS baseline matrix shows substantial off-diagonal mass. Of 175 true Level 1 cases, only 14 are recovered; 36 are predicted Level 2 (Orange) and 94 Level 3 (Yellow). Under-triage of critical presentations therefore dominates residual error on this US survey set.

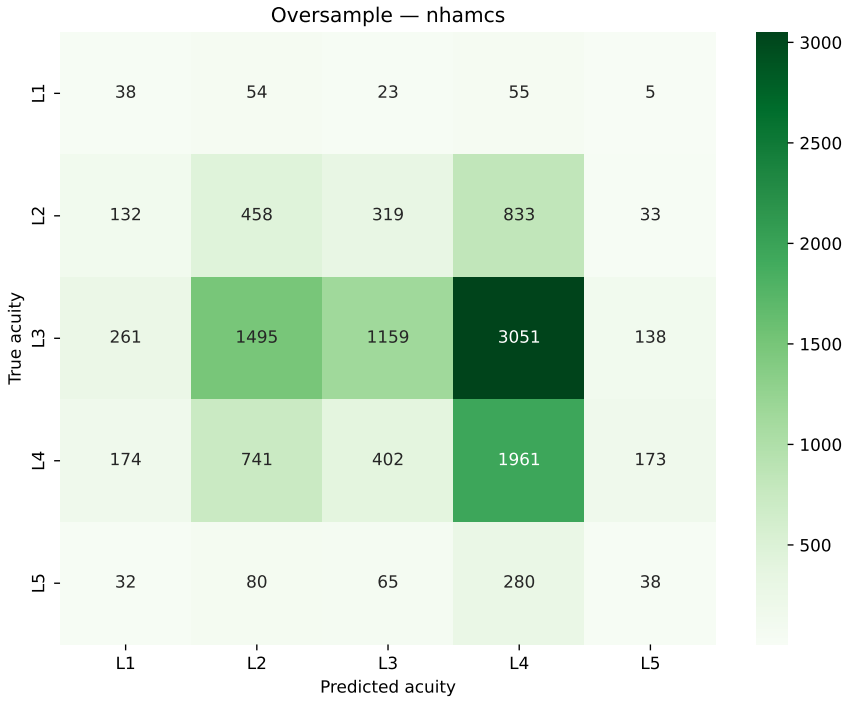


Figure 4.5: Stratified-oversampling BioBERT confusion matrix on the same NHAMCS subsample. Dark diagonal cells are correct predictions; off-diagonal mass is triage error.

Stratified oversampling raises NHAMCS Level 1 recall from 8.00% to 21.71% but lowers overall accuracy to 30.45% and Level 1 precision to 5.97%: more false urgent alerts accompany the sensitivity gain. As visually demonstrated by the drop in the Accuracy bars in Figure 4.2, stratified oversampling trades overall exact-match accuracy to raise Level 1 recall (Table 4.2). The baseline checkpoint remains the project default unless stakeholders explicitly prioritise Red sensitivity over precision (Berkowitz et al., 2019).

4.2.7 MIMIC-IV-ED demo results

On the open MIMIC-IV-ED demo ($N = 207$), baseline accuracy is 30.43% with colour accuracy 30.43% and high under-triage (63.77%). As shown in the baseline matrix (Figure 4.6), the model correctly recovers only 2 of 18 true Level 1 cases. With stratified oversampling (Figure 4.7), this increases to 9 of 18 cases (50.00% recall), though the matrix shows a corresponding spread of false positives across Levels 2 and 3, and overall accuracy falls to 21.26%. These demo figures validate the evaluation pipeline on real Boston ED phrasing but are not substitutes for a full credentialed MIMIC-IV-ED v2.2 extract (Johnson et al., 2023a).

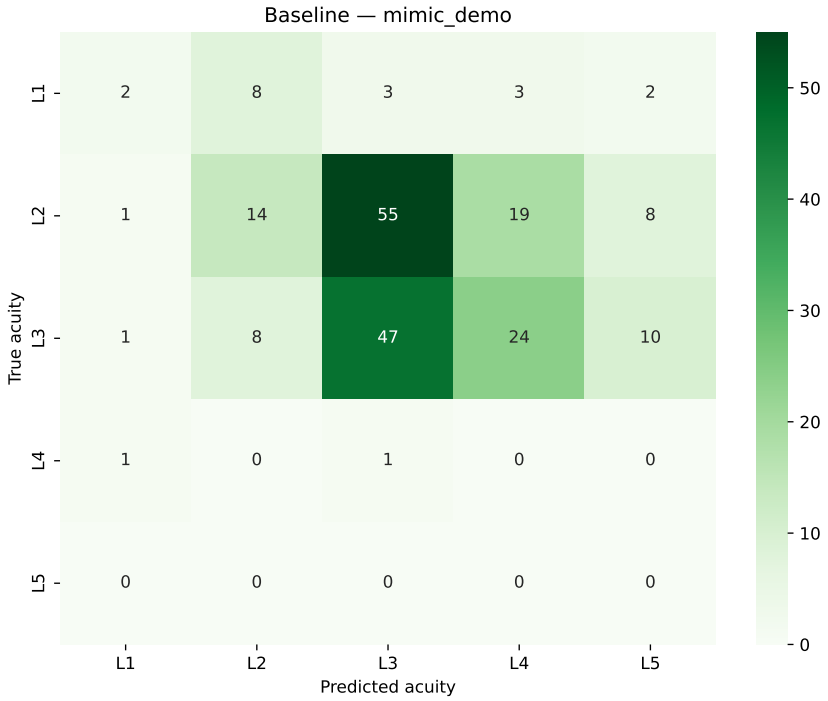


Figure 4.6: Baseline BioBERT confusion matrix on MIMIC-IV-ED demo ($N = 207$). Rows = true acuity; columns = prediction. Dark diagonal cells are correct predictions; off-diagonal mass is triage error.

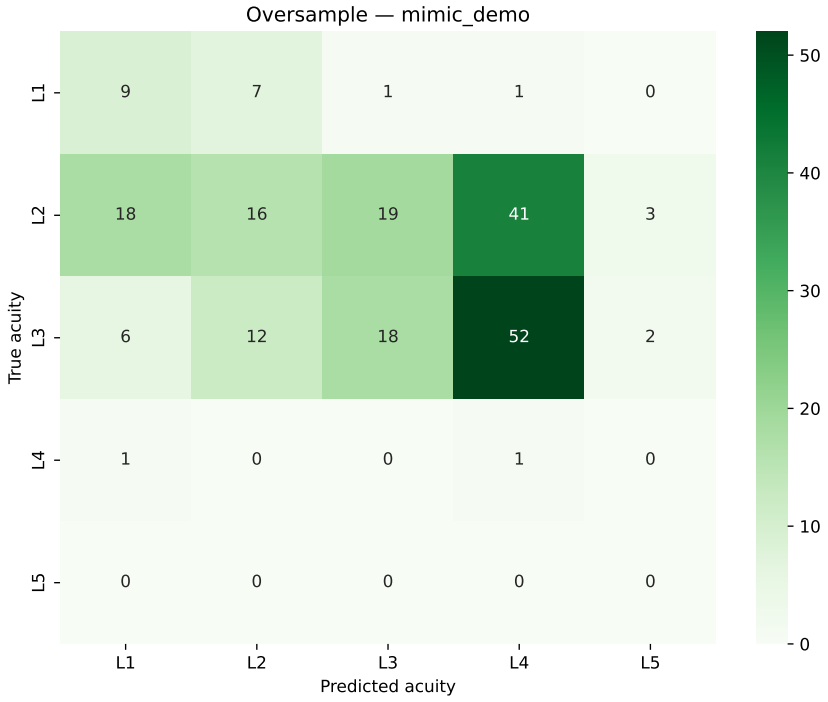


Figure 4.7: Stratified-oversampling BioBERT confusion matrix on MIMIC-IV-ED demo. Dark diagonal cells are correct predictions; off-diagonal mass is triage error.

4.2.8 Colour accuracy and under/over-triage

Mapping levels through f_{SATS} (Table 3.22) yields 4-colour accuracy at or above five-class accuracy when Levels 4 and 5 both map to Green. On NHAMCS baseline, colour accuracy (49.95%) slightly exceeds five-class accuracy (47.60%). Critical under-triage of true Red to non-Red remains high (92.00% baseline NHAMCS; 78.29% oversampled), reinforcing mandatory nurse override for urgent language (Berkowitz et al., 2019).

4.2.9 Inference latency

Real-time intake requires sub-second response (Stewart et al., 2023). Latency in Table 4.2 is a property of the BioBERT forward pass at project length $M = 128$ on the evaluation hardware, not a dataset-specific accuracy score. The published timing benchmark is therefore the NHAMCS stratified subsample ($N = 12{,}000$), which is large enough for stable median and tail estimates: baseline median latency ≈ 94 ms (p95 ≈ 124 ms); the stratified-oversampling checkpoint is similar (p50 ≈ 96 ms). Both respond in well under one second, supporting chatbot intake before nurse vitals are recorded.

MIMIC-IV-ED demo columns in the same table show n/a for latency by design. That demo ($N = 207$) is used for external accuracy, colour accuracy, and under/over-triage only. A second latency estimate on so few batches would not change the operational sub-second claim and would overstate the precision of a short timing sample. The same checkpoints and the same $M = 128$ truncation are used on both externals, so the NHAMCS p50 already represents chatbot response time at intake.

Remark 4.1. *External metrics are substantially lower than earlier Triagegeist in-distribution figures, as expected under geographic and demographic domain shift (Chapter 5). Text still carries a partial acuity signal, but the chatbot must remain nurse-overridable.*

4.3 Pathway routing at KNUST University Hospital

The complete five-scenario pathway map (gate reject plus Red, Orange, Yellow, Green) is specified in Tables 3.24 and 3.25 and illustrated in Figure 3.10. Each scenario assigns primary clinical departments, support actions across Nursing, Pharmacy, Health Information, Hospital Information Technology, Administration, and Accounts, and an escalation rule (South African Triage Group, 2012; Rominski et al., 2014).

4.3.1 Nurse triage vs chatbot response

Table 4.5 contrasts manual nurse triage with the hospital chatbot on dimensions relevant to intake at KNUST University Hospital. Nurse-side timings are drawn from published LMIC triage literature; chatbot timings are measured on the external evaluation hardware.

Dimension	Manual nurse triage	Hospital chatbot (this project)
Time to first acuity signal	Queue-dependent; LMIC intake often delayed minutes to hours before nurse contact (KATH Emergency Department, 2018; Mitchell et al., 2023; Nurchis et al., 2025)	Approximately 94–100 ms inference on evaluation hardware (Table 4.2)
Inputs used	Chief complaint + vitals + TEWS + clinical judgment (South African Triage Group, 2012)	Text-only chief complaint; TEWS applied separately by nurse (Section 3.3)
Output	SATS colour + bedside assessment	SATS colour + pathway card + confidence score
Validation basis	Live clinical decision at triage desk	External offline agreement on MIMIC/NHAMCS text (Section 4.2.2)
Role at KNUST University Hospital	Authoritative triage	Triage assist at arrival; nurse override mandatory for low confidence and all vitals (Stewart et al., 2023)

Table 4.5: Manual nurse triage versus chatbot assist at intake.

Once the chatbot assigns a colour in approximately 100 ms, pathway timer rules take effect: Orange cases require clinician contact within ten minutes, Yellow within 60 minutes, and Green within four hours (Section 3.14, Table 3.24) (South African Triage Group, 2012; Sundström et al., 2014). The chatbot does not replace nurse assessment; it is intended to provide an immediate acuity signal that could reduce queue time before *any* routing decision relative to waiting for a free triage nurse (Mitchell et al., 2023; Nurchis et al., 2025).

Gate rejection. Non-clinical input receives no colour; Nursing re-prompts for a chief complaint. This prevents meaningless acuity scores and keeps acute queues free of idle interactions.

Red. Life-threatening text (e.g. crushing chest pain with dyspnoea) routes immediately to Surgery and Internal Medicine with Code Red alerts via HIT. Administration bypasses routine registration so resuscitation is not delayed (South African Triage Group, 2012).

Orange. Very urgent complaints (e.g. severe abdominal pain in pregnancy) activate Internal Medicine, Surgery, or Obstetrics and Gynaecology within ten minutes, with stat

Diagnostics and a head-nurse escalation if unclaimed.

Yellow. For the running illustrative complaint (*"I have a headache and feel feverish. My body aches and I feel weak."*), the model assigns level 3 at approximately 72% confidence. The pathway card routes to the Outpatient urgent stream and parallel Diagnostics under the adult ASTS default; Pediatrics applies only under the prototype lexical child-marker rule (Section 3.14), which is not a reliable paediatric identifier. Nursing sets a 60-minute reassessment timer.

Green. Under the proposed routing, routine presentations (e.g. dental pain without fever) are intended to divert to Outpatient, Dental, Eye, or Public and Preventive Health within four hours, which may support reduced congestion in acute waiting areas (Yilmaz, 2021; Sundström et al., 2014).

Classification output (colour + confidence + department list) is the machine-readable input to this routing layer. External evaluation (Section 4.2.2) bounds offline reliability under domain shift; live intake at KNUST University Hospital still requires nurse override.

4.4 Discussion: anticipated impact on healthcare delivery

The following discussion interprets how the chatbot *could* support hospital flow at KNUST University Hospital, drawing on external metrics where available and on published triage literature elsewhere. These paragraphs are not measured outcome results. No prospective waiting-time trial has yet been conducted at KNUST University Hospital (Mitchell et al., 2023).

4.4.1 Functional chatbot and workflow improvement

We anticipate a functional chatbot that suggests ESI-style urgency from a single chief complaint and supports hospital workflow by recommending a SATS-aligned colour and department direction at intake (Stewart et al., 2023; Mitchell et al., 2023). External NHAMCS evaluation reports 47.60% five-class accuracy and 49.95% colour accuracy for the baseline checkpoint (Section 4.2.2), indicating that text carries a partial acuity signal under domain shift rather than near-perfect agreement. Median inference latency of approximately 94–100 ms on evaluation hardware supports rapid digital response at arrival (Table 4.5).

Figure 4.8 compares *time to computational acuity signal*: traditional intake (top) waits for a nurse before any colour exists; chatbot-assisted intake (bottom) produces a model colour recommendation in tens of milliseconds while nurse vitals (TEWS) still run in

parallel. The figure does *not* claim that total time from arrival to definitive clinical assessment equals inference latency.

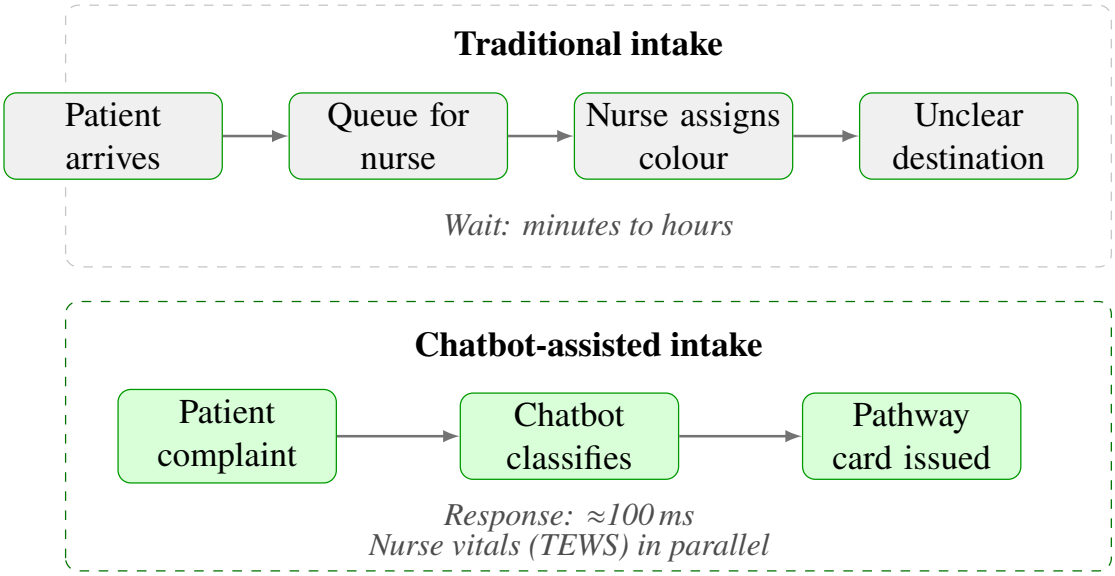


Figure 4.8: Time to computational acuity signal at KNUST University Hospital (schematic). **Top:** traditional intake may wait minutes to hours before a nurse assigns colour. **Bottom:** chatbot issues an ESI-style / f_{SATS} colour recommendation in ≈100 ms inference time; nurse TEWS vitals continue in parallel. This is *not* a measured reduction in total door-to-provider waiting time. Colour routing (Red/Orange vs Green) is detailed in Section 4.3.

The stacked workflow diagram has two panels read left to right. The top panel is traditional intake (arrive → queue → nurse assigns colour → unclear destination); the bottom panel is chatbot-assisted intake (complaint → chatbot classifies → pathway card issued). Pathway design comes from Chapter 3, chatbot latency from the external evaluation in Section 4.2.9, and traditional wait times from LMIC triage literature (KATH Emergency Department, 2018; Mitchell et al., 2023). The top footnote reads "Wait: minutes to hours"; the bottom footnotes read "Response: ≈100 ms" and "Nurse vitals (TEWS) in parallel". Red/Orange fast-track and Green digital queue are developed in Section 4.3. The figure therefore contrasts slow nurse-queue intake with immediate chatbot classification while making clear that nurse assessment is not replaced (South African Triage Group, 2012; Stewart et al., 2023).

4.4.2 Anticipated initial triage wait times

These paragraphs remain hypothetical. Published triage literature suggests that wait times for *initial* triage assessment *could* fall if patients interact with a bot immediately rather than queue solely for a nurse (Mitchell et al., 2023; Nurchis et al., 2025), but no door-to-provider trial was run here. External evaluation only shows that the model

can assign a colour and pathway recommendation in approximately 100 ms once valid clinical text is submitted (Table 4.5). Protocol design *intends* Green cases to be digitally queued and Red/Orange cases to be fast-tracked (Table 3.24); whether that shortens overall waiting duration at KNUST University Hospital is unmeasured (Sundström et al., 2014; Yilmaz, 2021).

4.4.3 Critical-case sensitivity and safety implications

The principal limitation of the baseline system is poor recall for the most urgent class under external domain shift. On NHAMCS, baseline Level 1 (Red) recall is only 8.00% (support 175): about 92% of true Red cases were not identified as Red, with most missed cases predicted as Orange or Yellow (Figure 4.4). Stratified oversampling raises Level 1 recall to 21.71% at large cost to precision, so no satisfactory operating point has yet been established for unsupervised critical-case screening. High offline sensitivity for critical cases is therefore *not* demonstrated on US external text; nurse override and prospective audit remain essential (Berkowitz et al., 2019). Waiting-time deterioration increases adverse outcomes when triage colour is not reassessed (Nurchis et al., 2025); digital timers on Orange and Yellow pathway cards are *intended* to reduce this risk, not proven to do so in this project (South African Triage Group, 2012).

4.4.4 Patient satisfaction and appropriate categorisation

Patient satisfaction and live clinical confirmation rates were not measured. A prospective pilot *could* ask whether quick pathway feedback improves perceived guidance (Stewart et al., 2023); offline NHAMCS/MIMIC agreement alone cannot answer that (Section 4.2.2).

4.4.5 Staff workload and public-health insights

Staff workload reduction and public-health surveillance benefits are likewise unmeasured hypotheses. Nursing documentation burden *might* shift if intake text capture is partly automated (Stewart et al., 2023; Mitchell et al., 2023); the gate can reject non-clinical chat before BioBERT (Section 3.15). Any future aggregation of anonymised complaints for trend analysis would require privacy and governance review first.

4.4.6 Success criteria

Prospective success at KNUST University Hospital would be judged by measured flow, nurse override rates, and adverse events or delays attributable to triage, not by offline Acc alone (Rominski et al., 2014; KATH Emergency Department, 2018). Those

criteria are targets for future evaluation, not results claimed in this thesis. **Limitations.** External metrics measure classification accuracy on US chief-complaint text, not patient outcomes, waiting-time reduction, or satisfaction scores at KNUST University Hospital (Mitchell et al., 2023; Nurchis et al., 2025). Prospective local validation is required. Training-data class imbalance and NHAMCS Level 1 scarcity widen uncertainty on rare urgent classes; deployment should treat low-confidence or contradictory presentations as mandatory nurse review (Berkowitz et al., 2019; South African Triage Group, 2012). Stratified oversampling increases Red recall on NHAMCS but sharply lowers Red precision; the baseline checkpoint is the project default unless clinical stakeholders prioritise sensitivity over precision.

4.5 Discussion of findings

Combining the model architecture analysis with the external metrics suggests five findings, addressing the research questions in Chapter 1.

Finding 1. BioBERT learns a partial acuity signal from free text after Stage 3 fine-tuning. Naive bounds remain near chance without that supervised step.

Finding 2. External domain shift substantially degrades performance relative to retired in-distribution Triagegeist headlines; NHAMCS baseline accuracy is 47.60% with ordinal MAE 0.627.

Finding 3. Class rebalancing (stratified oversampling) increases Red recall (8.00%→21.71% on NHAMCS) but introduces many false urgent predictions and worsens MAE, so no free lunch exists under $R(\theta)$.

Finding 4. The ESI→SATS map f_{SATS} is operational rather than clinically validated; colour accuracy is not interchangeable with five-class accuracy because Levels 4 and 5 share Green.

Finding 5. Latency near 94 ms is suitable for real-time interaction, but latency alone does not establish clinical utility or waiting-time reduction.

Safety posture. External five-class accuracy under domain shift does not license unsupervised discharge decisions. The deployed posture is triage assist with nurse override, TEWS in parallel, gate rejection of non-clinical text, and mandatory review when

$$\max_c \hat{y}_c < \tau_c \quad \text{or} \quad \hat{y}_{(1)} - \hat{y}_{(2)} < \delta$$

(with prototype $\tau_c = 0.60$, $\delta = 0.10$; Section 3.17), or when text conflicts with vitals (Stewart et al., 2023; Berkowitz et al., 2019). That posture is consistent with SATS training manuals and with under-triage quality literature.

CHAPTER 5

CONCLUSION

This project developed a mathematical pipeline from unstructured chief complaint text to an ESI-style acuity estimate, a SATS-aligned colour via f_{SATS}, and a protocol pathway card at hospital intake. The chatbot accepts a single English free-text submission and returns acuity probabilities, a colour recommendation, and a pathway card aligned with SATS practice and healthcare-facility department structure.

What was demonstrated.

1. A BioBERT classifier predicts ESI-style urgency from one English chief complaint. Primary reported metrics come from external evaluation on MIMIC-IV-ED and NHAMCS; Triagegeist synthetic pairs are used for training and validation only.

2. A deterministic heuristic f_{SATS} maps ESI-style levels to SATS-aligned colours; pathway cards $P(C_{\text{NLP}})$ name destination, $T_{\max}$, actions, and escalation, while TEWS remains the nurse vital-sign path in parallel.

3. A clinical gate with threshold 0.30 withholds colour from non-clinical chat before BioBERT runs, with Acc/Sens/Spec reported on a labeled contrast set (Section 3.15).

What the experiments showed. External evaluation indicates that free-text chief complaints carry a measurable but limited acuity signal under geographic domain shift. Training with additional Level 1 (Red) examples to offset class imbalance raises Red recall at a clear cost to overall accuracy and precision (RQ4). Softmax mode probability is a model score, not calibrated clinical confidence.

What was not demonstrated. This project did not demonstrate live Ghanaian clinical safety, prospective waiting-time reduction, patient-outcome improvement, or clinician-adjudicated SATS equivalence of f_{SATS}. Pathway tables reflect the authors' documented interpretation of observed workflows and published SATS rules; they are not an official hospital protocol unless formally approved.

Ultimately, this project shows that NLP can be mathematically constrained and operationalized for low-resource hospital settings as nurse-overridable decision support. By bridging unstructured patient narratives to deterministic SATS-aligned pathway

protocols at KNUST University Hospital, the Flow Management Engine provides an auditable specification for nurse-overridable intake routing; whether it reduces bottlenecks or waiting times at KNUST University Hospital remains for prospective evaluation.

5.1 Limitations and future work

Data scope. Triagegeist is used for Stage 3 training and validation only. Published classification metrics in Chapter 4 are computed on external MIMIC-IV-ED and NHAMCS text, after dropping empty complaints and retaining acuity labels in $\{1, \ldots, 5\}$, with tokenisation truncated to project length $M = 128$. The model is English-only at intake. An observational site visit to KNUST University Hospital mapped pathways and department structure through intake observation and staff inquiries; supplementary English example complaints generated from those protocols support prototype and qualitative analysis only and were not used for fine-tuning. A natural next step is prospective collection of locally labeled chief complaints at KNUST University Hospital for local retraining, and exploring optional fusion of vitals and TEWS alongside free text when measurements become available.

Label mismatch. Training and external gold labels are ESI-style; target pathway colours are SATS-aligned via f_{SATS}. These are related but not identical clinical systems.

Cross-domain and cross-linguistic shift. Formally, $P_{\text{train}}(X, Y) \neq P_{\text{Ghana}}(X, Y)$. Formal English training text differs from real Ghanaian patient phrasing (Pidgin, Twi-influenced English).

Geographic and demographic domain shift. The chatbot is designed for Ghanaian facility pathways at KNUST University Hospital, yet the external test corpora are United States sources. MIMIC-IV-ED is drawn from a Boston-area teaching hospital population that is predominantly White (approximately 68%), with smaller shares of Black (approximately 15%), Hispanic/Latino (approximately 5%), and Asian (approximately 3%) patients in published MIMIC demographic descriptions. NHAMCS likewise reflects a North American ambulatory and emergency survey population, not Ghanaian intake language mix, care pathways, or case-mix. External accuracy therefore does *not* equal expected live performance at KNUST University Hospital; prospective local validation remains essential.

Class imbalance and Level 1 scarcity. Real emergency departments are heavily skewed toward mid-acuity visits. NHAMCS in particular has very few Level 1 (Resuscitation) cases relative to Levels 3–4. Chapter 4 reports class support N_ℓ alongside recall so rare-class metrics are interpreted cautiously. Stratified oversampling on the Triagegeist training split studies the recall–precision trade-off; classical embedding-space SMOTE was not used for the published oversampled checkpoint (Section 4.2.5).

Age and pediatrics. The classifier is age-agnostic: it maps $X \rightarrow \hat{y}$, not $(X, \text{age}) \rightarrow \hat{y}$, and defaults to adult SATS (ASTS) TEWS and pathway tables. Pediatrics is not a default Yellow destination; the present destination override uses a prototype lexical child-marker rule only and can miss age cues such as "8-year-old" or "girl with . . . ". Structured age input is future work.

No clinical adjudication of colour. No clinician independently validated chatbot colours against SATS for each external test case.

No live workflow or outcome experiment. Waiting-time reduction and adverse-event rates were not measured.

No Softmax calibration study. Maximum Softmax probability is not established as calibrated clinical probability.

Clinical validation. The model has not yet been clinically tested on real patients at KNUST University Hospital. External offline evaluation suggests text carries a limited acuity signal under domain shift, but Softmax confidence should still trigger nurse oversight when $\max_c \hat{y}_c < \tau_c$, when the top-two margin $\hat{y}_{(1)} - \hat{y}_{(2)} < \delta$, or when language conflicts with vitals (Section 3.17). The clinical relevance gate was validated only on a small hand-labeled contrast set ($N = 52$), enough for a prototype threshold check but not for broad claims about live gate performance at KNUST University Hospital; lexicon-poor clinical wording can still produce false clinical rejections and must be handled by re-phrasing or nurse override (Section 3.15). The next phase is a prospective pilot at KNUST University Hospital to measure waiting times, nurse override rates, and patient outcomes in live intake (Mitchell et al., 2023).

Pathway refinement. The five-scenario pathway map is SATS-aligned and complete for gate reject, Red, Orange, Yellow, and Green routing, including Green diversion from acute queues. The pathway configuration represents the authors' documented interpretation of the observed workflow and should not be interpreted as an official clinical protocol unless formally approved by the hospital. Ongoing collaboration with hospital administrators and clinical leads at KNUST University Hospital will keep department routing tables current as service lines evolve (Rominski et al., 2014; South African Triage Group, 2012).

Language and equity. The current model processes formal English. Expanding the system to handle Ghanaian informal English (Pidgin) or integrating voice-to-text capabilities for Twi would significantly broaden accessibility for patients who are less comfortable writing formal English complaints.

System integration. The prototype chatbot API and pathway cards already anchor the chatbot in a deployable pipeline. Deeper integration with Hospital Information Technology and Health Information systems would automate timer alerts and bed assignment, completing the Flow Management Engine vision across Administration, Diagnostics, and all clinical departments listed in Table 3.26.

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Appendix A

Software repository

The implemented prototype and fine-tuned BioBERT artefacts are maintained at:
`https://github.com/samuel6814/triage`
No full source code is reproduced in this project; the mathematical specification in Chapter 3 and the system architecture in Figure 3.12 are the primary references.

Appendix B

Prototype `/predict` response fields

A successful clinical submission returns a JSON object whose fields mirror the mathematics of Chapter 3. Table B.1 lists the principal keys.

Field	Meaning
`gate_score`	Clinical relevance $g(X)$ in $[0, 1]$
`gate_passed`	True if $g(X) \geq 0.30$
`probabilities`	Softmax vector $\hat{\mathbf{y}}$ over levels 1–5
`predicted_level`	$\hat{c} = \arg\max_c \hat{y}_c$
`sats_colour`	$C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$
`pathway`	Pathway card $P(C)$: $T_{\max}$, destination, actions, escalation
`confidence`	Mode probability $\hat{y}_{\hat{c}}$ for nurse display

Table B.1: Principal `/predict` response fields and their mathematical counterparts.

Gate failures omit colour and pathway fields and instead return a prompt requesting a clinical chief complaint.

Appendix C

Stage 3 fine-tuning checklist

For reproducibility, Stage 3 training used the following checklist (full values in Table 3.20):

1. Join Triagegeist `train.csv` and `chief_complaints.csv` on `patient_id`.
2. Keep only the English free-text chief complaint (why the patient came) and the ESI-style urgency label (1–5); drop vitals and demographics.
3. Stratified 90/10 split with seed 42 ($N_{\text{train}} \approx 72{,}000$, $N_{\text{eval}} = 8{,}000$).
4. Load BioBERT encoder; discard any prior two-class head; initialise $W \in \mathbb{R}^{5\times 768}$, $\mathbf{b} \in \mathbb{R}^5$.
5. Train three epochs, batch size 16, learning rate 2×10^{-5}, weight decay 0.01, max length 128, FP16 enabled.
6. Select the checkpoint with best validation cross-entropy (0.001812 at epoch 3 for the baseline run).
7. Optional stratified-oversampling branch: augment and randomly oversample minority urgent classes on the training split before the same training recipe; compare on external MIMIC/NHAMCS sets (not a Triagegeist holdout headline).

Appendix D

Glossary of principal symbols

Symbol	Definition
X	English chief-complaint string
τ	WordPiece tokenisation map
M	Sequence length (maximum 128)
D	Hidden dimension (768)
$H^{(\ell)}$	Hidden-state matrix after encoder layer ℓ
$h_{[\mathrm{CLS}]}$	Summary vector (row 0 of $H^{(12)}$)
$W, \mathbf{b}$	Classification weight matrix and bias
$\mathbf{z}, \hat{\mathbf{y}}$	Logits and Softmax (Categorical) probabilities
$\hat{c}$	Predicted acuity level
f_{SATS}	Level-to-colour map
C_{NLP}	NLP-assigned SATS colour
$P(C)$	Pathway card for colour C
$g(X)$	Clinical relevance gate score
T	TEWS integer from vitals

Table D.1: Compact glossary of symbols used across Chapters 3–4.